



Remote sensing image enhancement and water eutrophication prediction based on atmospheric-water multimodal information fusion

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Abstract

Irregular temporal sampling and limited data availability restrict the performance of remote sensing-based chlorophyll-a prediction for water eutrophication assessment. Meanwhile, atmospheric pollutant deposition represents an important pathway for nutrient input into lakes, yet such information is rarely incorporated into data-driven prediction frameworks. To address these challenges, this study proposes a multimodal fusion framework integrating atmospheric pollutant data and chlorophyll remote sensing imagery. First, a transfer learning-based generative adversarial network (TL-MCAS-CSG-GAN) is developed to reconstruct missing chlorophyll remote sensing images and enhance temporal continuity. Second, a cross-attention Conv2D-LSTM GAN model is designed to fuse atmospheric six-parameter features with chlorophyll image sequences for eutrophication prediction. Experiments conducted on Taihu Lake datasets demonstrate that the proposed model improves SSIM, PSNR, and COSIN by 6.51% compared with baseline models without atmospheric feature fusion. The results indicate that atmospheric pollutant information provides complementary spatiotemporal signals for chlorophyll prediction and improves eutrophication assessment accuracy. The proposed framework offers a data-driven paradigm for multimodal environmental modeling and contributes to intelligent water quality monitoring and management.

Keywords Remote sensing images · Graph data · Image migration expansion · Multi-modal fusion · Eutrophication prediction

Introduction

Water eutrophication refers to the pollution phenomenon caused by excessive content of nutrients such as nitrogen, phosphorus, and organic matter in water bodies with low internal circulation (Lin et al. 2021; Petrova et al. 2023). It typically occurs in areas where terrestrial and

aquatic ecosystems intersect (Hennings et al. 2021; Politi et al. 2024), such as lakes and rivers, which are strongly affected by human activities and have poor water exchange capabilities (Dai et al. 2023). In recent years, with the acceleration of urbanization and industrialization, an increasing amount of pollutants has been discharged (Madjar et al. 2024; Lin et al. 2022), making water eutrophication a major challenge in global water ecological governance (Baladi et al. 2022; Preisner et al. 2020). Since the 1960s, the global incidence of eutrophication has been increasing. From the late 1970s to the late 1990s, the proportion of eutrophic lakes rose from 41% to 61% (Zhang et al. 2021). In 2019, among the 107 lakes (reservoirs) monitored in China, 5.6% were classified as moderately eutrophic, 23.4% as mildly eutrophic, 61.7% as mesotrophic, and 9.3% as oligotrophic (Yu et al. 2020). These findings indicate an overall concerning water quality status. Therefore, accurate prediction of water body eutrophication is of great significance (Zang et al. 2024).

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Eutrophication prediction studies focus on applying mathematical models and remote sensing techniques to foresee its occurrence and development trends. These models are based on the physical and chemical characteristics of water bodies, biological indicators, and atmospheric conditions. Advances in remote sensing technology have improved the efficiency of large-scale water body monitoring and early warning (Laneve et al. 2024; Peppas et al. 2020). Prediction methods can be divided into two categories based on whether they reveal the intrinsic mechanisms of water bodies: Mechanism-driven prediction focuses on direct and indirect influencing factors, describing their relationship with eutrophication through mathematical equations. This approach has evolved from single-nutrient limitation models (Du et al. 2022; Zhang et al. 2023) to incorporating temperature variations (He et al. 2024) and suspended solids (Pan et al. 2023), and further to complex models integrating ecosystem dynamics and hydrodynamic processes (Koue 2025). Such progression has deepened the multidimensional understanding of eutrophication and improved prediction accuracy. On the other hand, data-driven prediction models use big data and data mining techniques to analyze observational information, combining ground stations and remote sensing data. These models cover both specific water quality details at individual sites and broad water body characteristics, revealing key influencing factors and patterns through data analysis and providing important decision-making support for water environment management (Chen et al. 2021; Karimian et al. 2023; Li et al. 2021; Zhang et al. 2022). These two methods complement each other and jointly promote the scientific prediction and management of water eutrophication.

In addition to watershed nutrient inputs, atmospheric deposition has been increasingly recognized as an important external nutrient source influencing eutrophication processes. In large shallow lakes such as Taihu Lake, atmospheric nitrogen deposition has been shown to contribute significantly to nutrient loading and algal bloom dynamics (Chen et al. 2018). Existing studies generally adopt two main approaches to incorporate atmospheric factors into eutrophication research. The first approach introduces atmospheric nitrogen deposition and related meteorological variables into water quality models as external driving forces. Modeling studies have demonstrated that atmospheric nitrogen deposition can measurably affect water quality dynamics and nutrient concentrations (Wang et al. 2022). The second approach couples climate forcing with hydrodynamic–ecological models to simulate the influence of atmospheric processes on nutrient transport and eutrophication evolution. Recent research has shown that atmospheric deposition and climate warming jointly alter water quality responses and ecosystem dynamics (Han et al.

2023). These findings indicate that atmospheric processes are not merely background environmental variables but can act as external nutrient drivers influencing chlorophyll-a variability. Therefore, integrating atmospheric pollutant information into chlorophyll-a prediction models is theoretically grounded and scientifically justified. In this study, the atmospheric dataset consists of daily PM_{2.5} images with 1 km spatial resolution, while chlorophyll-a data are derived from satellite inversion products with irregular temporal sampling. This difference in temporal resolution further motivates the need for temporal reconstruction and multimodal modeling.

Convolutional neural networks, as a classic deep learning algorithm, overcome the curse of dimensionality in high-dimensional data analysis and are particularly suitable for image processing and recognition. Convolutional processing makes feature extraction from images more efficient, providing a more streamlined input set for subsequent nonlinear transformations, thereby improving training and prediction effects (Archana and Jeevaraj 2024; Liang 2024; Liu et al. 2024; Zhao et al. 2025). Generative adversarial networks (GANs), as excellent models for generating image data, consist of a generator and a discriminator. The generator learns to produce samples that approximate the distribution of real remote sensing images, while the discriminator distinguishes whether the images come from real remote sensing image samples or are generated by the generator. This allows the network model to effectively extract key features from water body remote sensing images and generate high-quality water body remote sensing image data (Gui et al. 2021; Jabbar et al. 2021; Pai et al. 2020).

To address the aforementioned challenges, this study develops a multimodal fusion framework integrating atmospheric pollutant information and chlorophyll remote sensing imagery for water eutrophication prediction. The main contributions of this work are summarized as follows:

- (1) A transfer learning-based image reconstruction framework (TL-MCAS-CSG-GAN) is proposed to address irregular temporal sampling and limited chlorophyll data availability, improving temporal continuity of remote sensing image sequences.
- (2) A cross-attention multimodal fusion architecture is designed to integrate atmospheric six-parameter features with chlorophyll image sequences, enabling joint spatiotemporal modeling across heterogeneous environmental data sources.
- (3) A structured ablation analysis is conducted to quantitatively evaluate the contribution of temporal modeling, attention mechanisms, and atmospheric feature fusion, demonstrating the effectiveness of multimodal integration for chlorophyll-a prediction.

- (4) The proposed framework provides a data-driven paradigm for integrating atmospheric and water quality information in eutrophication assessment, contributing to intelligent environmental monitoring and Earth system informatics research.

Water body remote sensing image expansion method based on atmospheric pollutant knowledge transfer

Considering that chlorophyll is the main indicator for predicting water eutrophication, this paper uses chlorophyll remote sensing images as the dataset for modeling. However, the original chlorophyll remote sensing image data of the target water body has unequal sampling lengths and low sampling frequencies, resulting in unequal time intervals and a small volume of data samples, making the original data unsuitable for direct modeling and prediction. Therefore, it is necessary to study a method for expanding the chlorophyll remote sensing image samples of the target water body. Additionally, since the volume of chlorophyll remote sensing images is small, expanding these samples requires rich prior knowledge. Traditional methods based on single factors for expanding chlorophyll remote sensing image samples often produce results that differ significantly from real images. Considering the strong correlation between atmospheric pollutants and chlorophyll concentration, it is necessary to

study a method for expanding chlorophyll remote sensing image samples based on high-resolution atmospheric pollutant image data as prior knowledge. In response to these issues, this chapter proposes a chlorophyll remote sensing image expansion method based on transfer learning and multi-channel attention cascaded semantic selection networks.

The model for expanding water body remote sensing images based on atmospheric pollutant knowledge transfer consists of four modules, as shown in Fig. 1.

The first module is the cascaded semantic generation module, which overlays atmospheric pollutant images with the semantic maps of the target water body and uses them as inputs to the model. This module extracts features from atmospheric pollutant images and generates coarse-grained chlorophyll remote sensing images of the target water body. The second module is the multi-scale spatial pooling module, which performs average pooling on the coarse-grained cascaded semantic features at different scales to expand the receptive field and obtain feature information from a larger area. The third module is the multi-channel attention selection module, which convolves multi-scale features and combines attention mechanisms to generate more accurate results. The fourth module is the knowledge transfer module, which integrates the generated chlorophyll remote sensing images with the original chlorophyll remote sensing image data and adjusts the generated images using the original data to achieve smooth fusion of the expanded image dataset.

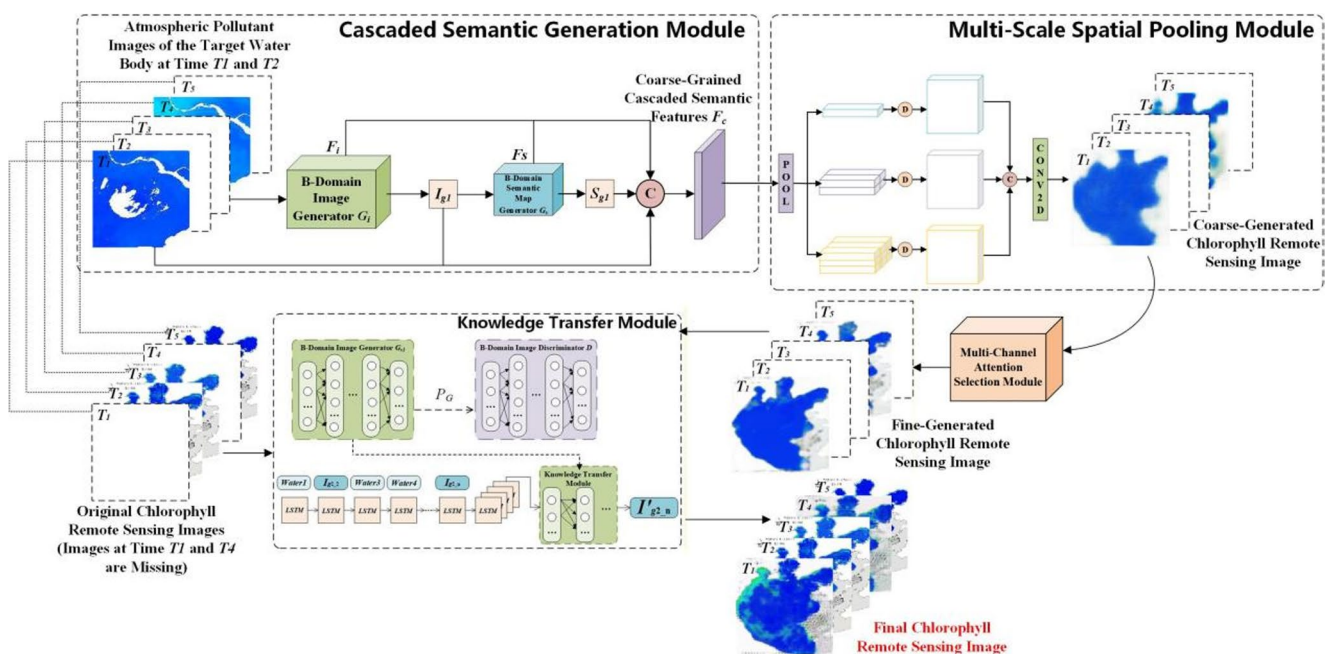


Fig. 1 Structure diagram of water remote sensing image expansion method based on atmospheric pollutant knowledge transfer

Construction of the cascaded semantic generation module

The task of image translation and generation across different knowledge domains is a challenging endeavor, particularly when the domains represent different physical quantities. This study investigates how to leverage knowledge learned from atmospheric pollutant images of a target water body to generate reference chlorophyll remote sensing images. To enhance the accuracy of image translation, this paper employs semantic maps of the target water body as conditional guidance. A U-Net-based semantic segmentation model is employed to extract semantic maps of the target water body. However, using semantic mappings solely in the reconstruction loss to guide the generation of chlorophyll remote sensing images provides only weak guidance for image generation. Therefore, this study not only incorporates the semantic maps of the target water body in the reconstruction loss but also uses them as input to the model. The specific cascaded semantic generation module structure is illustrated in Fig. 2, where the image I_a from domain A and the semantic map S_g from domain B are concatenated through function $\text{concat}(\bullet)$ and fed into the image generator G_i of domain B, producing the output image I_{g1} of domain B.

In this way, the semantic map S_g from domain B provides stronger supervision for image generation in subsequent units, formalized as:

$$I_{g1} = G_i(\text{concat}(I_a, S_g)) \quad (2.1)$$

Considering the cascaded target water body chlorophyll images and semantic maps can enhance the accuracy of the model's generated images. Therefore, this paper constructs a cyclic semantic generation network to enable the model

to better learn image feature information. Specifically, the output image I_g from the image generator G_i is further fed into the semantic map generator G_s of domain B to construct a new semantic map S_{g1} . This provides additional informational guidance for the generator, allowing it to better learn the information from the images of domain A and the semantic maps of domain B. This process is formalized as:

$$S_{g1} = G_s(I_{g1}) = G_s(G_i(\text{concat}(I_a, S_g))) \quad (2.2)$$

Due to the complexity of the task, it is observed that after the initial step, the image generator G_i of domain B produces a coarse-grained target domain image I_{g1} . The generated image exhibits blurriness, with a notable disparity in high pixel-level similarity between the scene details and the target domain image, as illustrated in Fig. 3.

This necessitates exploring a coarse-to-fine image generation strategy to improve image generation accuracy. Cascaded processing of images has been applied in various computer vision tasks, such as object detection and semantic segmentation, and has demonstrated significant effectiveness. In this section, a cascaded strategy is introduced to address the problem of image translation and generation across different knowledge domains. The output of the last convolutional layer of the image generator G_i of domain B is extracted as the coarse-grained target domain image feature F_i . Similarly, the output of the last convolutional layer of the semantic map generator G_s of domain B is extracted as the coarse-grained semantic map feature F_s . The image I_a from domain A, the coarse-grained target domain image I_{g1} , the target domain semantic map S_{g1} , the feature F_i , and the feature F_s are then cascaded to obtain the new feature F_c , as formalized below.

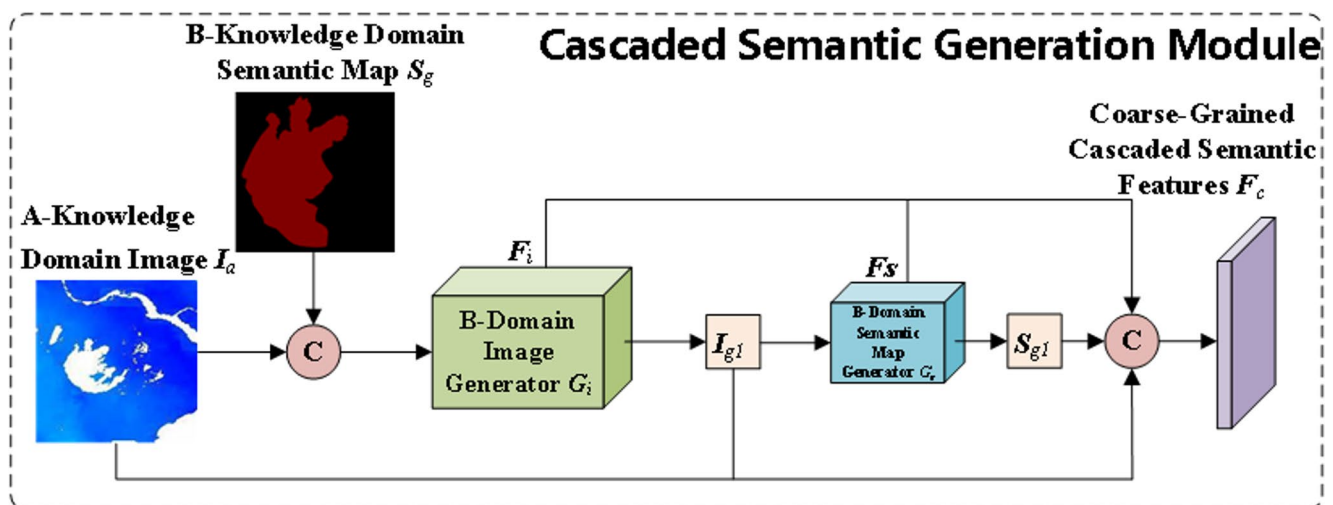


Fig. 2 Structure diagram of cascaded semantic generation module

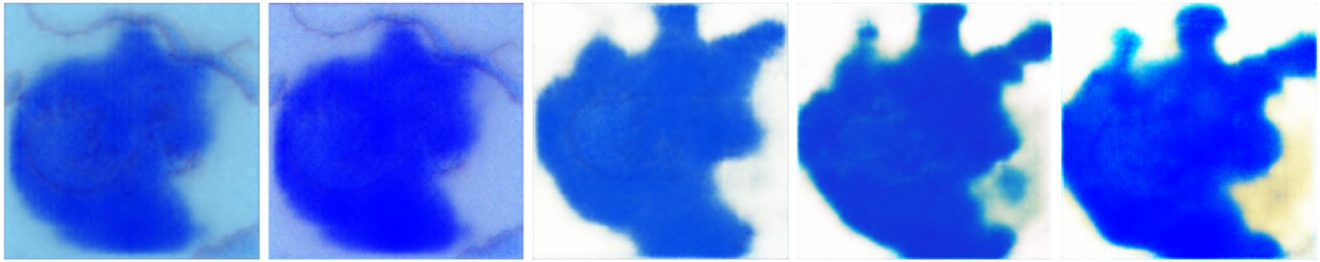


Fig. 3 Example of coarse-grained target domain image

$$F_c = \text{concat}(I_a, I_{g1}, S_{g1}, F_i, F_s) \quad (2.3)$$

The concatenation operation integrates complementary information from different sources. The original atmospheric image preserves spatial distribution patterns, while the coarse-generated image and semantic maps provide structural constraints for chlorophyll reconstruction. Channel-wise concatenation allows the model to retain diverse feature information before further refinement. Although redundant information may be introduced, subsequent multi-scale pooling and attention mechanisms automatically suppress irrelevant features and enhance useful representations.

In the implementation example of this paper, the water quality remote sensing image of the monitored target water body is missing. For each missing moment, the atmospheric pollutant image of that moment is obtained, and then the atmospheric pollutant image and the semantic map of the target water body are input into the cascaded semantic generation module. The module generates a coarse-grained water quality remote sensing image, from which feature F_c is extracted.

Since the PM2.5 dataset has daily temporal resolution, temporal alignment was performed based on calendar-day matching. For each chlorophyll observation (or missing timestamp), the atmospheric image from the same date was selected.

Construction of the multi-scale spatial pooling module

To fully utilize the coarse-grained feature F_c output by the cascaded semantic generation module, this section proposes a multi-scale spatial pooling module. Due to significant variations in objects and scenes between images from domain A and domain B, single-scale feature extraction cannot capture all the spatial information required for fine-grained image generation. Therefore, this section introduces a multi-scale spatial pooling module, which applies global average pooling to the input features using a set of M kernels with different sizes but the same stride. Specifically, a set of M spatial scales $\{s_i\}_{i=1}^M$ is used in the pooling process,

resulting in pooled features at different spatial scales. These features are then concatenated along the channel dimension and passed through a Conv2d convolutional layer to obtain the multi-scale feature F_{c1} . The model structure is illustrated in Fig. 4, and the process is formalized as follows:

$$F_{c1} \leftarrow \text{concat}(F_c \times \text{up}_1(F_c), \dots, F_c \times \text{pl_up}_M(F_c)) \quad (2.4)$$

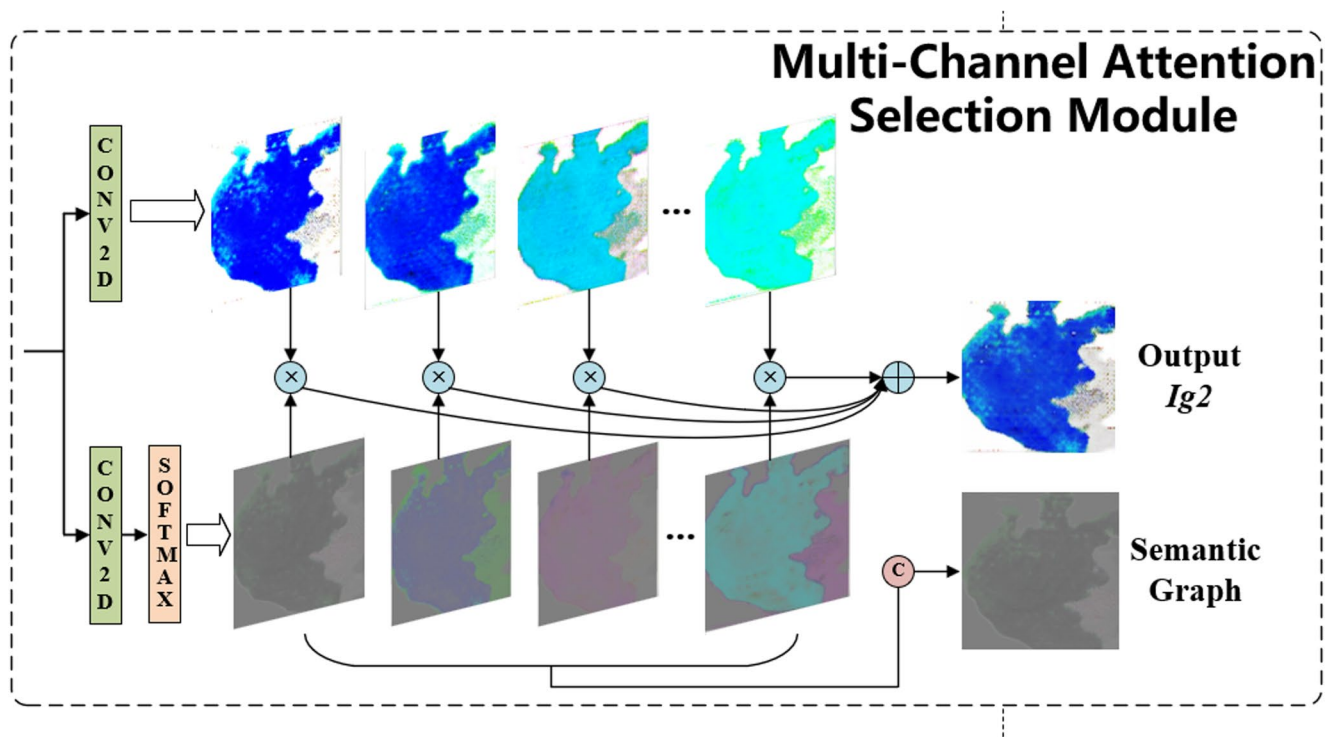
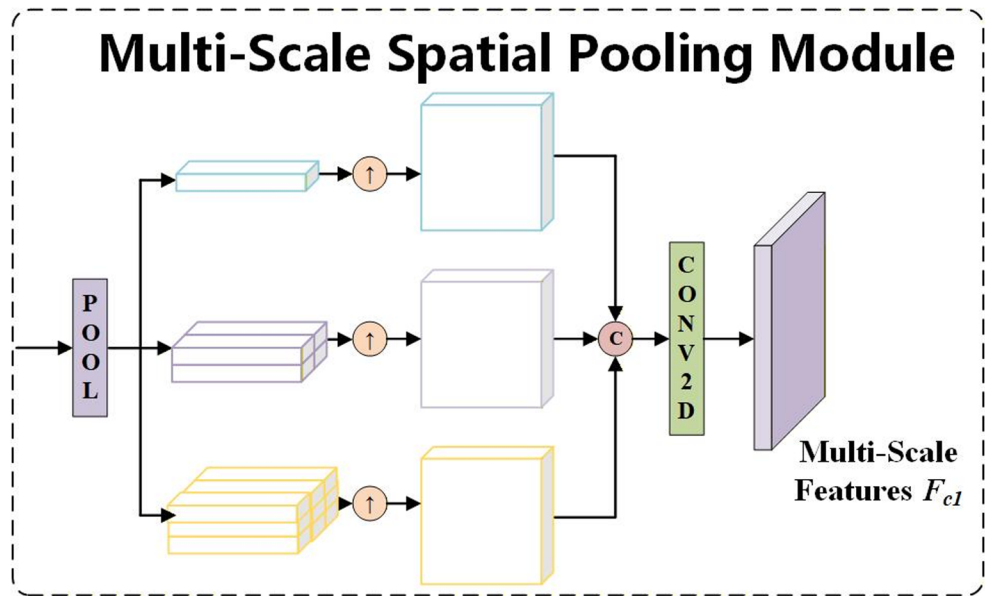
where up_i denotes the upsampling operation.

Construction of the multi-channel attention selection module

In the first two unit modules, outputs are generated in a three-channel RGB representation format corresponding to the color-encoded chlorophyll concentration maps. However, for cross-domain image conversion generation in this study, this is limiting. It is necessary to use a larger generation space to create multiple intermediate variable images to achieve richer synthesis. Therefore, this paper designs a multi-channel attention mechanism that automatically selects spatially and temporally from the generation to synthesize a finer final output. The model structure is illustrated in Fig. 5.

Given a multi-scale feature volume $F_{c1} \in R^{h \times w \times c}$, where h and w represent the width and height of the features, respectively, and c is the number of channels, this section designs two image output streams. One is used to generate intermediate variable images for multiple B knowledge domains, and the other is used to generate multiple channel attention maps. To generate N different intermediate variables $I_G = \{I_G^i\}_{i=1}^N$, N convolutional filters $\{W_G^i, b_G^i\}_{i=1}^N$ are first applied for convolution operations, followed by a nonlinear activation operation $\tanh(\cdot)$. For generating the corresponding N attention maps, another set of filters $\{W_A^i, b_A^i\}_{i=1}^N$ is applied. The formulas for calculating the intermediate variable images of multiple B knowledge domains and the multiple channel attention maps are as follows:

$$I_G^i = \tanh(F_{c1} W_G^i + b_G^i), \text{ for } i = 1, \dots, N \quad (2.5)$$

Fig. 4 Structure diagram of multi-scale spatial pooling module**Fig. 5** Structure diagram of multichannel attention module

$$I_A^i = \text{Softmax} \left(F_{cl} W_A^i + b_A^i \right), \text{ for } i = 1, \dots, N \quad (2.6)$$

Here, $\text{Softmax}(\cdot)$ is the activation function used to normalize the multiple channel attention maps. Finally, the learned attention maps are applied to perform channel-wise selection from the intermediate variables of each generator, as follows:

$$I_g' = (I_A^1 \times I_G^1) + \dots + (I_A^N \times I_G^N) \quad (2.7)$$

where I_g^N represents the final synthesized output selected from multiple different results, and the symbol $+$ denotes element-wise addition. In the third module, the final semantic map of the B knowledge domain is obtained. This paper establishes the following relationship:

However, the semantic maps obtained from the pre-trained model are not accurate, which leads to erroneous guidance during the training process. To address this issue, this paper proposes using the generated attention maps to learn uncertainty maps, thereby controlling the optimization loss. This approach is introduced to tackle the problem of noisy semantic labels, assuming that there are K different loss maps requiring guidance. First, multiple generated attention maps are concatenated and then passed through a convolutional layer with K filters $\{W_u^i\}_{i=1}^K$ to produce a set of K uncertainty maps. The reason for using attention maps to generate uncertainty maps is that the attention maps directly influence the final generated results and have a strong connection with the loss. Let L_p^i represent the pixel-level loss map, and U_i denote the i -th uncertainty map. The relationship is as follows:

$$U_i = \sigma(W_u^i(\text{concat}(I_A^1, \dots, I_A^N)) + b_u^i) \quad (2.8)$$

$$L_p^i \leftarrow \frac{L_p^i}{U_i} + \log U_i, \text{ for } i = 1, \dots, K \quad (2.9)$$

where $\sigma(\cdot)$ is the *Sigmoid* function used for pixel-level normalization. The uncertainty maps are learned automatically and serve as a weighting scheme to control the optimization loss.

Construction of the knowledge transfer module

The specific structure of the knowledge transfer module is illustrated in Fig. 6. First, a generative adversarial network (GAN) is used as the feature extractor. The B-domain image generator G_{s1} learns the features of chlorophyll remote sensing images from the original target water body, while the B-domain image discriminator D completes adversarial training to uncover the rules for generating chlorophyll remote sensing images. Additionally, the B-domain image generator G_{s1} can eliminate redundant information. The original water quality remote sensing images are input into the feature extractor for learning. After learning the rules for generating chlorophyll remote sensing images, the network structure and parameters of the generator G_{s1} are transferred and cascaded with an LSTM network, to form the knowledge transfer module. This module models the spatial and temporal correlations of chlorophyll remote sensing image data.

The expanded water quality remote sensing image sequence, consisting of both generated and original observations arranged in chronological order, is fed into the LSTM network. Through its gating mechanism, the LSTM captures long-term temporal dependencies between consecutive time steps and mitigates abrupt variations potentially introduced by the generative process. As a result, the module outputs a temporally smoothed sequence of chlorophyll remote sensing images.

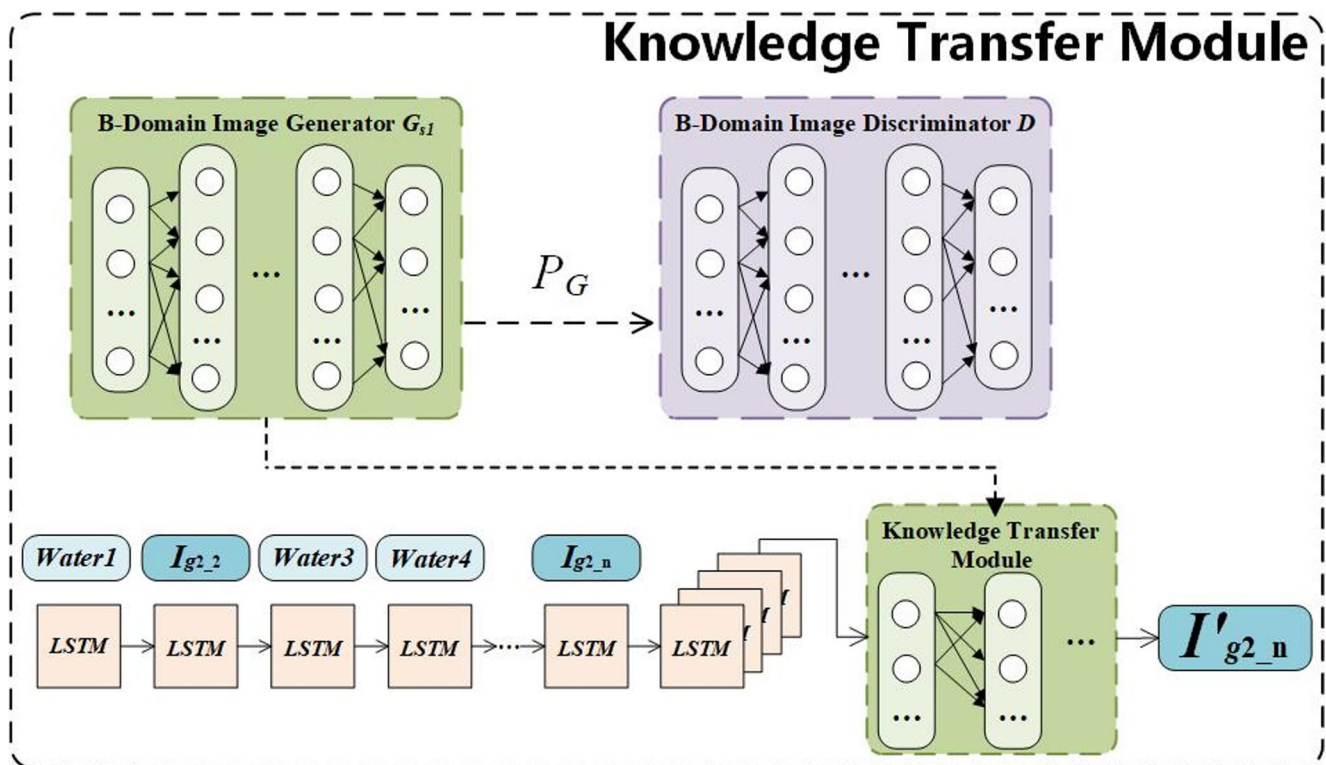


Fig. 6 Structure diagram of knowledge transfer module

Water eutrophication prediction method based on multimodal information fusion

Construction of the spatiotemporal feature extraction module

The structure of the atmospheric pollutant station time series data feature extraction module is shown in Fig. 7. Considering that the time series data of atmospheric pollutant stations in multiple regions have high-dimensional, spatiotemporal evolution, and multifactor characteristics, a four-dimensional directed GCN-LSTM model is constructed. The four-dimensional directed graph convolutional unit can include two-dimensional spatial directional information and capture the correlation information of multiple factors within graph nodes. Additionally, the traditional four-dimensional directed graph convolutional filter is improved and optimized to enhance the model's data processing efficiency. Then, by introducing the LSTM network to extract temporal correlation information at different time points, a four-dimensional directed GCN-LSTM model is constructed to synchronously

extract spatiotemporal information. The atmospheric pollutant image sequence is input into the four-dimensional directed GCN-LSTM model to extract image features. In the example shown in Fig. 7, each batch inputs an atmospheric pollutant image sequence with a time length of 30, with 6 atmospheric pollutant concentrations at 7 spatial nodes. The four-dimensional directed GCN-LSTM model extracts spatiotemporal features F_{air} at 30 time points, with a feature dimension of D2.

The structure of the chlorophyll remote sensing image time series data feature extraction module is shown in Fig. 8. The Att_Conv2D_LSTM module has unique advantages in processing time series image data, as it can fully utilize both the spatial and temporal information of the images (Lin et al. 2020). The input to this module is historical water quality remote sensing image data. First, the time series image data is normalized to the same size scale, and the sequence data is organized according to the time steps. The front end consists of multiple convolutional layers that extract local features such as edges and textures from the images. Then, the series of feature maps output after convolutional layer processing are passed through an attention

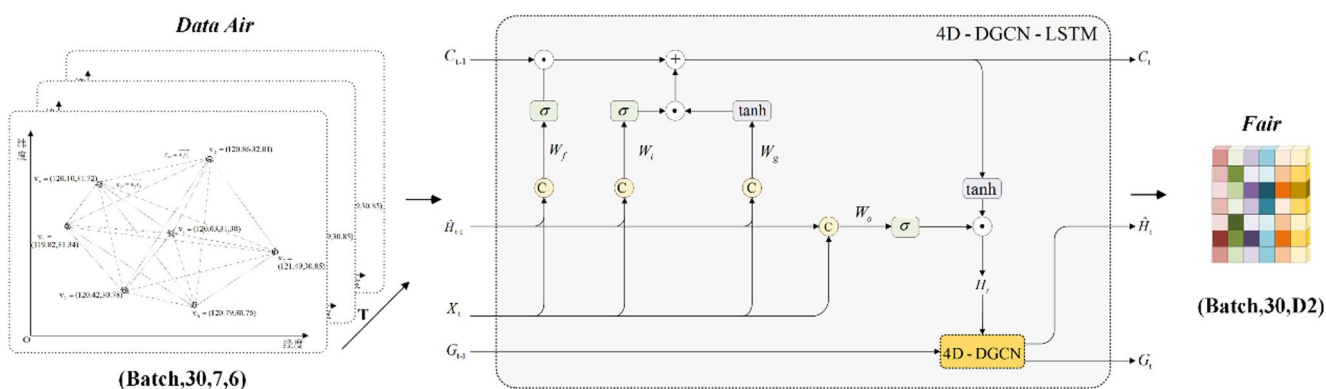


Fig. 7 Structure diagram of atmospheric pollutant station time series data feature extraction module

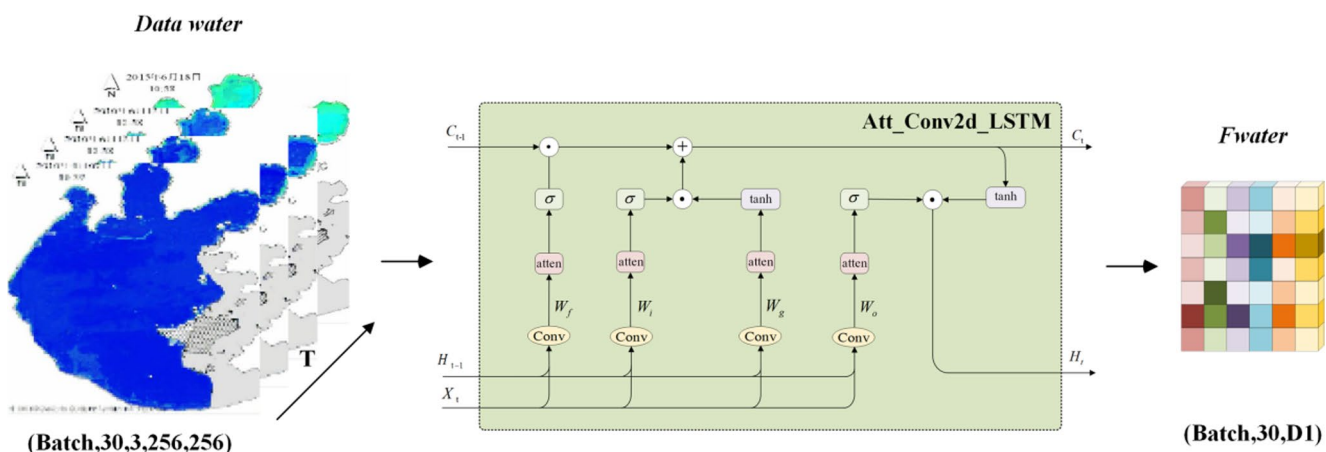


Fig. 8 Structure diagram of Chlorophyll remote sensing image time series data feature extraction module

mechanism to retain spatial information while reducing data dimensionality. Finally, the transformed feature vectors are input into the LSTM layer to capture temporal dependencies in the sequence data. Through the gating mechanism (input gate, forget gate, and output gate), information is selectively retained and transmitted, enabling the model to learn temporal features in the image sequence. In the example shown in Fig. 8, Each batch of input chlorophyll concentration sequences has a temporal length of 30, represented in a three-channel image format (256×256 pixels) corresponding to the color-encoded concentration maps. After processing by the Att_Conv2D_LSTM model, spatiotemporal features F of the size $D1$ at 30 time points are obtained.

Construction of the single-modal feature encoder

The spatiotemporal features of atmospheric pollutant images and the spatiotemporal features of target water body remote sensing images are input into the single-modal feature encoder for feature dimension alignment.

The single-modal feature encoder consists of a self-attention layer and a feedforward network layer, as shown in Fig. 9. Among them, the self-attention layer can automatically calculate the weights of different positions in the input sequence, enabling the encoder to focus more on the features most relevant to the current task. This helps the model filter out the most useful information from a large amount of data (Hui et al. 2023; Wang et al. 2024). The feedforward network layer can learn nonlinear transformations, further converting and extracting the feature representations output by the self-attention module, aligning the feature dimensions of different modal data. In this paper, the atmospheric six-parameter and chlorophyll remote sensing image spatiotemporal features F_{air} and F_{water} obtained from the feature extraction module are input into the single-modal feature encoder, and the output is the atmospheric six-parameter feature F_{air_sa} and chlorophyll remote sensing image feature F_{water_sa} with the same dimensions.

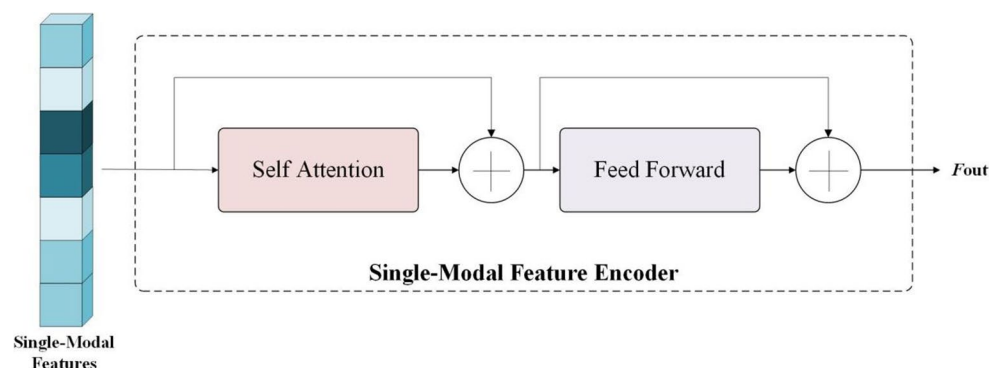
Construction of the cross-modal feature encoder

The cross-modal feature encoder is composed of a bidirectional cross-attention layer (Cross Attention), a self-attention layer (Self Attention), and a feedforward network (Feed Forward), as shown in Fig. 10. Among them, the cross-attention layer is primarily used for information interaction between different modalities, allowing data from one modality to focus on and acquire feature information from another modality, thereby achieving feature complementarity and enhancement across modalities (Li et al. 2022; Luo et al. 2024). The self-attention layer is mainly used to focus on information interaction within a modality, capturing internal dependencies within the modality and improving the overall representation capability of the encoder. The feedforward network performs nonlinear transformations on the features output by the attention layers to extract higher-level abstract features. In this paper, the outputs F_{air_sa} and F_{water_sa} of the single-modal feature encoder are input into the cross-modal feature encoder, resulting in chlorophyll remote sensing image features $F_{water_sa_air}$ that incorporate atmospheric six-parameter factors.

Generation and discrimination module

The generation and discrimination module consists of a Markov discriminator and a feature decoder. The Markov discriminator is a special convolutional neural network structure composed of Conv2d convolutional layers, LeakyReLU activation function layers, and Sigmoid activation function layers, as shown in Fig. 11. The convolutional layers use a set of learnable filters to slide over the input image, performing weighted summation on local regions to obtain new feature representations, capturing local patterns and structural information in the image, and providing a basis for subsequent discrimination or generation tasks. The LeakyReLU activation function layer is used to increase the nonlinear capability of the neural network, helping to prevent gradient vanishing and improve the training effect of the model. The Sigmoid activation function layer maps the output of the

Fig. 9 Structure diagram of single-modal feature encoder



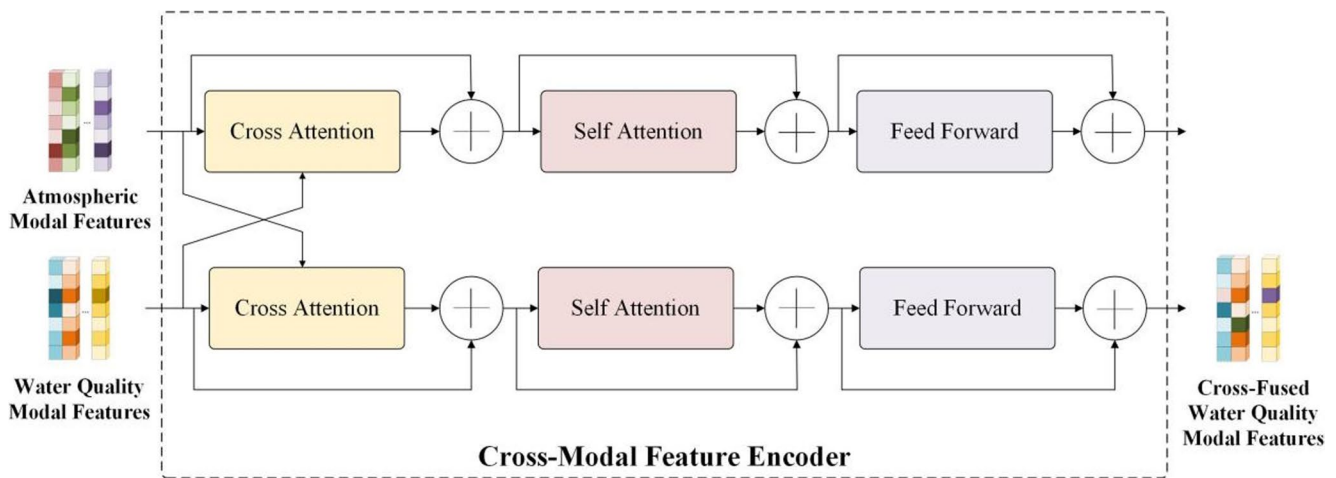


Fig. 10 Structure diagram of cross-modal feature encoder

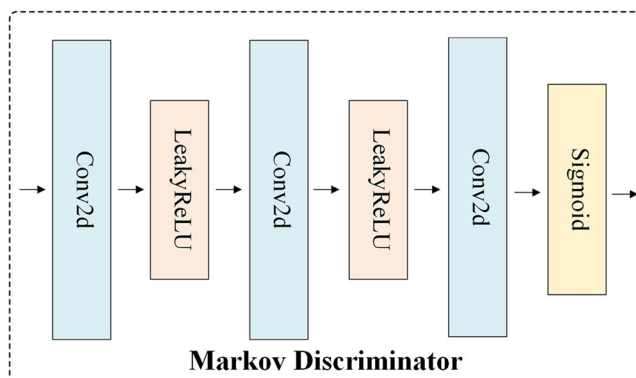
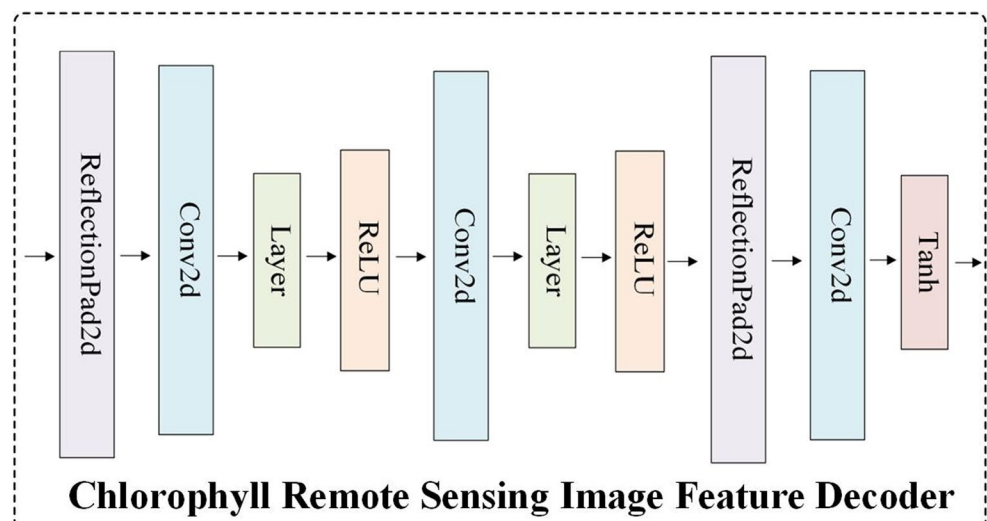


Fig. 11 Structure diagram of markov discriminator

network to a range of 0 to 1, providing the probability value of the correctness of the discrimination. Finally, the network outputs a feature map corresponding to the local regions of the input image, where each feature value reflects the probability that the region conforms to the real data distribution.

Fig. 12 Structure diagram of chlorophyll remote sensing image feature decoder



The feature decoder consists of a ReflectionPad2d boundary padding function, Conv2d convolutional layers, Layer linear layers, ReLU activation functions, and Tanh activation functions, as shown in Fig. 12. ReflectionPad2d is a two-dimensional reflection padding layer in PyTorch that performs reflection padding on the edges of the input feature map to protect edge information and avoid the loss of edge information during convolution operations. The Conv2d convolutional layer mainly performs weighted summation on local regions of the input image by sliding a convolution kernel to obtain new feature representations, effectively extracting local features and spatial relationships in the image, which is crucial for image recognition and processing tasks. Additionally, convolutional operations have the characteristics of weight sharing and local perception, which can reduce the number of model parameters and improve computational efficiency. The Layer linear layer is typically used for linear transformation of input features, mapping the output of the previous layer to the input of the next layer. Each neuron in the linear layer

is connected to all neurons in the previous layer, performing linear combination through weights and biases, enabling the learning and adaptation to different data distributions and task requirements. The ReLU function has the characteristic of unilateral suppression, which can enhance the nonlinear characteristics of the neural network, helping the network learn features better. At the same time, the ReLU function is computationally simple, which can speed up model training. Moreover, the ReLU function maintains the gradient unchanged when the input is positive, helping to alleviate the gradient vanishing problem. The Tanh function has a wide output range, which can better handle negative inputs. This makes it very effective in processing images with complex textures and structures. At the same time, the Tanh function has the effect of suppressing interference, helping the neural network filter out noise and irrelevant information.

Experimental verification

To avoid circular validation, we strictly separate reconstructed (generated) samples from the evaluation set. The reconstructed chlorophyll-a images are used only for training to enhance temporal continuity, while all quantitative evaluations are conducted exclusively on real chlorophyll-a observations. The dataset is split chronologically into training, validation, and test subsets to prevent future information leakage. Metrics such as SSIM, PSNR, and COSIN are computed on the test set consisting only of real observations, and generated samples are excluded from testing.

Experimental data

Chlorophyll-a concentration is widely regarded as a key indicator for water eutrophication assessment. In this study, chlorophyll remote sensing images are used as the primary dataset for modeling. The chlorophyll-a data are provided as color-encoded inversion maps, where each RGB value corresponds to a specific chlorophyll concentration defined by the original data product. However, the original chlorophyll observations of the target water body are characterized by irregular temporal sampling and limited data volume, resulting in discontinuous time series that are not suitable for direct spatiotemporal prediction.

To address this issue, an image expansion strategy is introduced to enhance temporal continuity. Given the limited number of chlorophyll samples, auxiliary environmental information is incorporated to provide additional contextual support during reconstruction. Atmospheric pollutant data, which may be associated with nutrient deposition processes influencing eutrophication dynamics, are therefore used as complementary prior information.

The experimental datasets adopted in this study are summarized in Table 1.

Among them, the sample of the Yangtze River Delta 1 km high-resolution daily PM_{2.5} data is shown in Fig. 13. The sample of Taihu Lake chlorophyll remote sensing image data is shown in Fig. 14. The atmospheric six-parameter dataset for the Taihu Lake urban agglomeration is presented, showing the spatial-temporal samples of PM_{2.5} (Fig. 15), PM₁₀ (Fig. 16), CO (Fig. 17), O₃ (Fig. 18), NO₂ (Fig. 19), SO₂ (Fig. 20). The sample of the expanded Taihu Lake chlorophyll remote sensing image dataset is shown in Fig. 21.

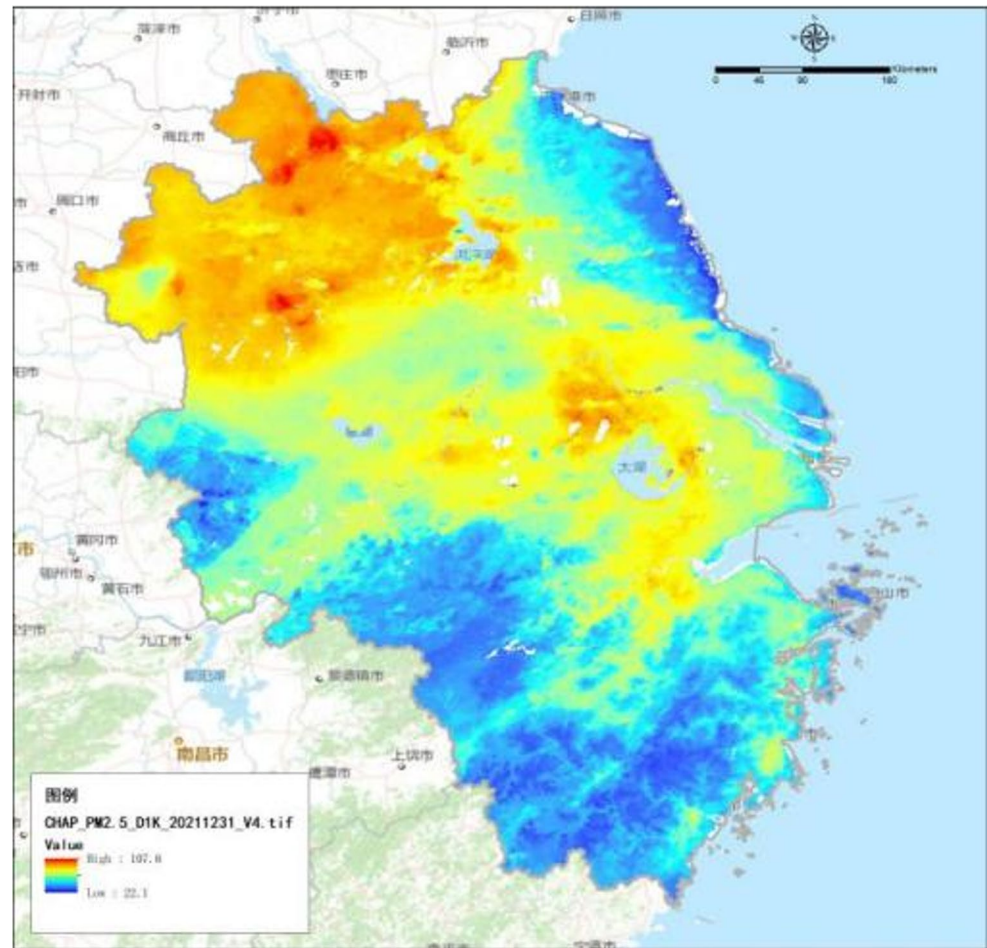
Construction and prediction results of the TL-MCAS_CSG_GAN model

Considering that the Yangtze River Delta PM_{2.5} high-resolution image data is sampled once a day, while the Taihu Lake chlorophyll remote sensing image data is missing

Table 1 Experimental datasets used in this paper

Name	Source	Introduction	Data size
Yangtze River Delta 1 km High-Resolution Daily PM _{2.5} Dataset	National Earth System Science Data Center	Geographic scope: Jiangsu, Zhejiang, and Anhui provinces in the Yangtze River Delta region; Time resolution: daily; Variable unit: ug/m ³ ; Data format: Geo-TIFF files	Time length: 2014 to 2017; Sample size: 1461 images
Taihu Lake Chlorophyll Remote Sensing Image Dataset	National Earth System Science Data Center	Geographic scope: Taihu Lake Basin; Time resolution: irregular; Variable unit: ug/L; Data format: jpg images	Time length: 2014 to 2017; Sample size: 569 images
Atmospheric Six-Parameter Dataset of Taihu Lake Urban Agglomeration	WheatA Agricultural Meteorological Big Data	Geographic scope: Suzhou, Nantong, Changzhou, Yixing, Huzhou, Jiaxing, and Shanghai; Time resolution: 3 h; Variable content: PM _{2.5} , PM ₁₀ , O ₃ , CO, SO ₂ , NO ₂	Time length: 2014 to 2017; Sample size: 490,896 records
Taihu Lake Chlorophyll Remote Sensing Image Expansion Dataset	National Earth System Science Data Center and Experimental Generation	Geographic scope: Taihu Lake Basin; Time resolution: daily; Variable unit: ug/L; Data format: jpg images	Time length: 2014 to 2017; Sample size: 1461 images

Fig. 13 Sample of chlorophyll-a remote sensing image data for Taihu Lake



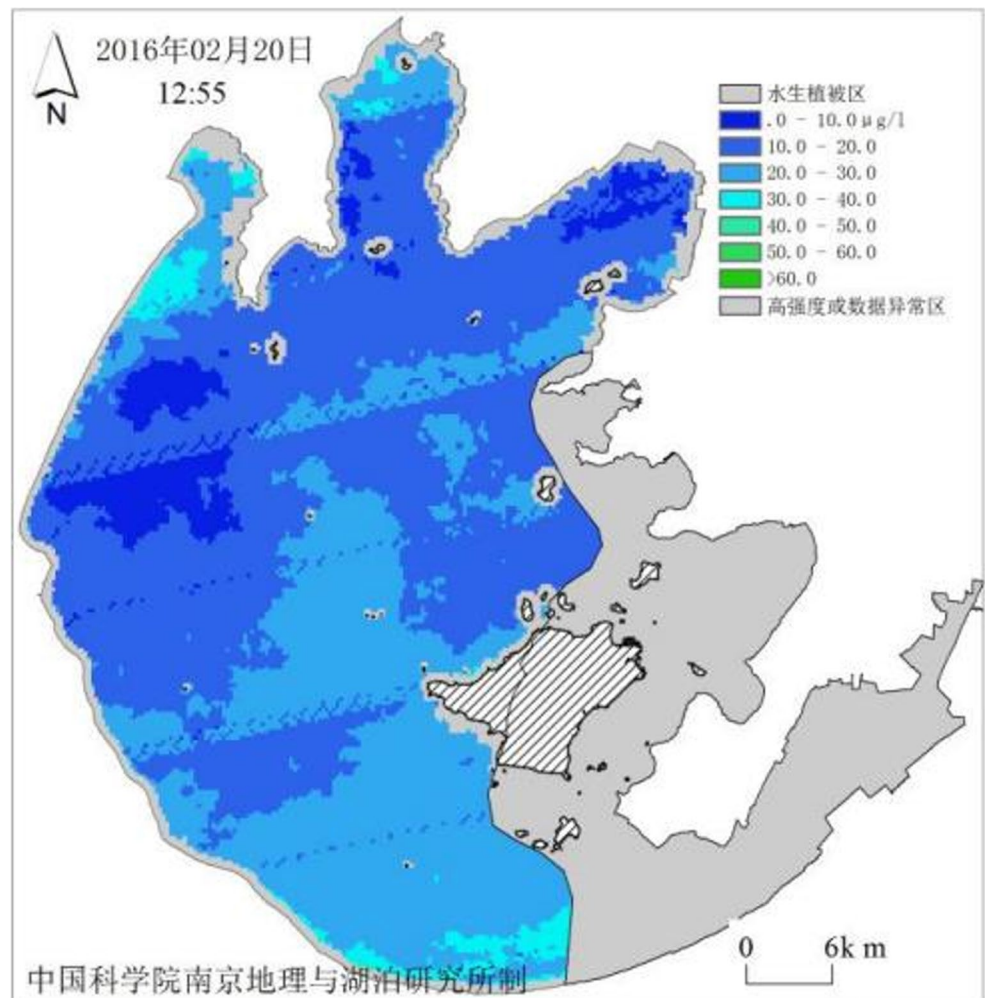
without a temporal pattern, the following explains the experimental process of the Taihu Lake remote sensing image sample expansion model based on atmospheric pollutant knowledge transfer. The missing conditions of the existing two datasets are shown in Table 2, Fig. 22, and Fig. 23.

This section uses the Zhixing Cloud computing platform to support experimental calculations, selecting the Windows 10 operating system, equipped with a dual-core GPU, graphics card type GeForce RTX 3090 (24GB video memory), system image win10 cuda11, system disk size 350GB, and memory size starting from 56GB. The deep learning environment is configured with cuda11.1, cuDnn8, Anaconda, Pytorch1.7.1, Tensorflow2.4, and python3.8.3. In network training, the initial learning rate is set to 0.01, the batch size is set to 64, and the overall training cycle is set to 150 epochs. Additionally, the Adam optimizer is selected for its adaptive learning rate adjustment characteristics, which helps the model optimize the loss function more effectively during training. The specific experimental process of the model is shown in Table 3:

This section compares the effects of four models (GAN, CSG_GAN, MCAS_GAN, MCAS_CSG_GAN) and the TL-MCAS_CSG_GAN model (the proposed model in this study) in expanding and generating 5 missing image data in 15 consecutive time points. For fairness, all comparative models were trained and evaluated under the same experimental settings. The missing data generation process followed an identical strategy across models, where temporally adjacent observations were used as input to predict the target time step. The training data, preprocessing procedures, and loss functions were kept consistent to ensure that performance differences were solely attributed to architectural variations rather than experimental bias. The experimental results are shown in Figs. 30, 31, 32, 33, and 34, where Fig. 28 shows a complete time series of Taihu Lake chlorophyll remote sensing image data, and Fig. 29 shows a set of manually missing time series of Taihu Lake chlorophyll remote sensing image data with 5 images removed.

The purpose of the experiment is to verify that the proposed TL-MCAS_CSG_GAN model is the most effective

Fig. 14 Sample of 1 km high-resolution daily PM2.5 data in the Yangtze River Delta



model for expanding Taihu Lake chlorophyll remote sensing images compared to the four models (GAN, CSG_GAN, MCAS_GAN, MCAS_CSG_GAN). By comparing the real results in Fig. 28 with the experimental results in Figs. 30, 31, 32, 33, and 34, and analyzing the evaluation metrics of each model in Table 4, the following conclusions can be drawn:

Comparing the three evaluation metrics of the GAN model and the CSG_GAN model, the SSIM, PSNR, and COSIN values of the GAN model are 0.61547, 19.32871, and 0.87164, respectively, while the SSIM, PSNR, and COSIN values of the CSG_GAN model are 0.63701, 19.90857, and 0.89953, respectively, representing improvements of 0.02154, 0.57986, and 0.02789, with an average improvement percentage of 3.12%. It can be concluded that the cascaded semantic generation unit can gradually construct and enrich semantic information. By cascading multiple semantic generation units, each unit can further process and optimize based on the output of the previous unit, learning and fusing various semantic information and

generation strategies, thereby improving the overall image generation quality.

Comparing the three evaluation metrics of the GAN model and the MCAS_GAN model, the SSIM, PSNR, and COSIN values of the MCAS_GAN model are 0.64070, 20.14051, and 0.90563, respectively, representing improvements of 0.02523, 0.8118, and 0.03399, with an average improvement percentage of 4.07%. It can be concluded that the multiscale spatial pooling unit can perform pooling operations at different spatial scales, extracting feature information at different scales, helping the model better capture details and global information in images or data. The multichannel attention selection unit can weight features based on the importance of different channels, making the model focus more on key features that contribute to the task, thereby enhancing the model's ability to capture key information and improving the richness and accuracy of features.

Comparing the three evaluation metrics of the CSG_GAN model, MCAS_GAN model, and MCAS_CSG_GAN model, the SSIM, PSNR, and COSIN values of

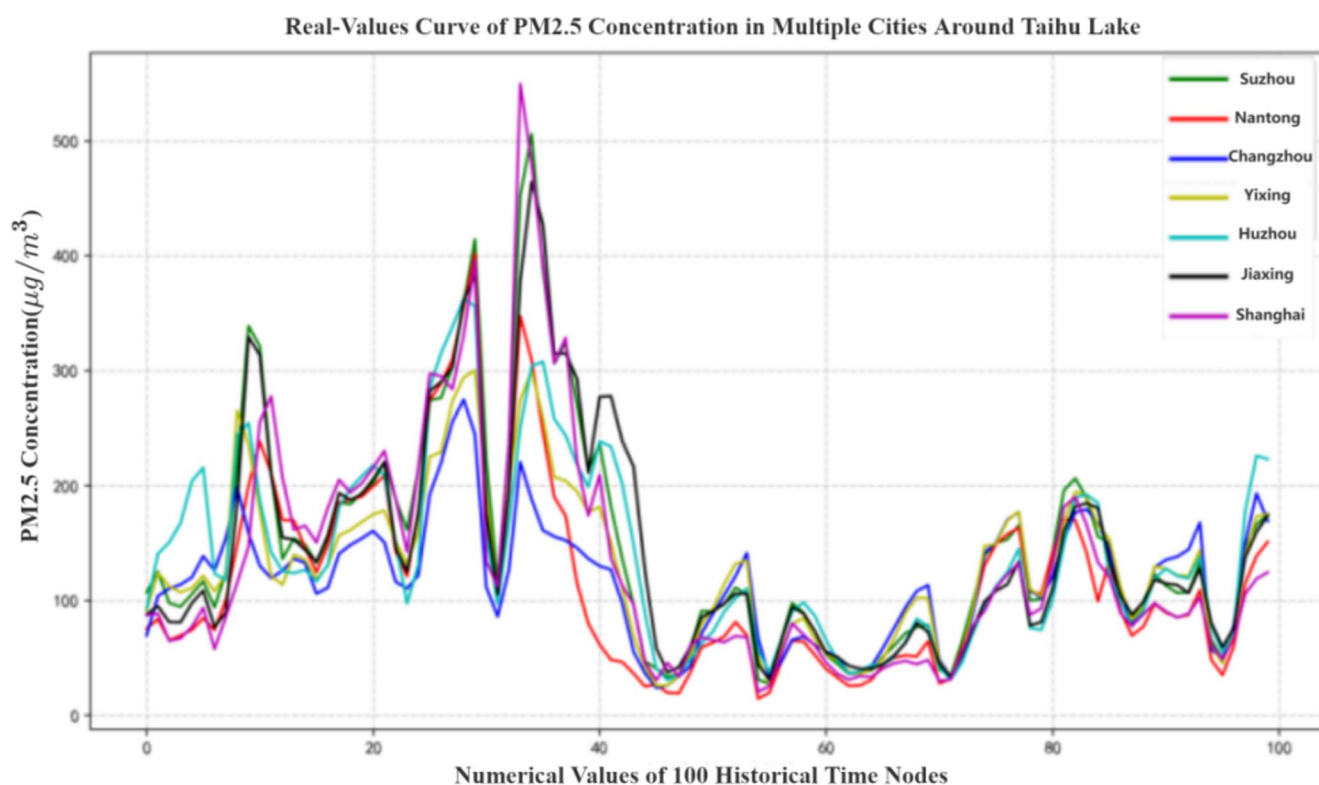


Fig. 15 Sample of PM2.5 concentration data for cities surrounding Taihu Lake

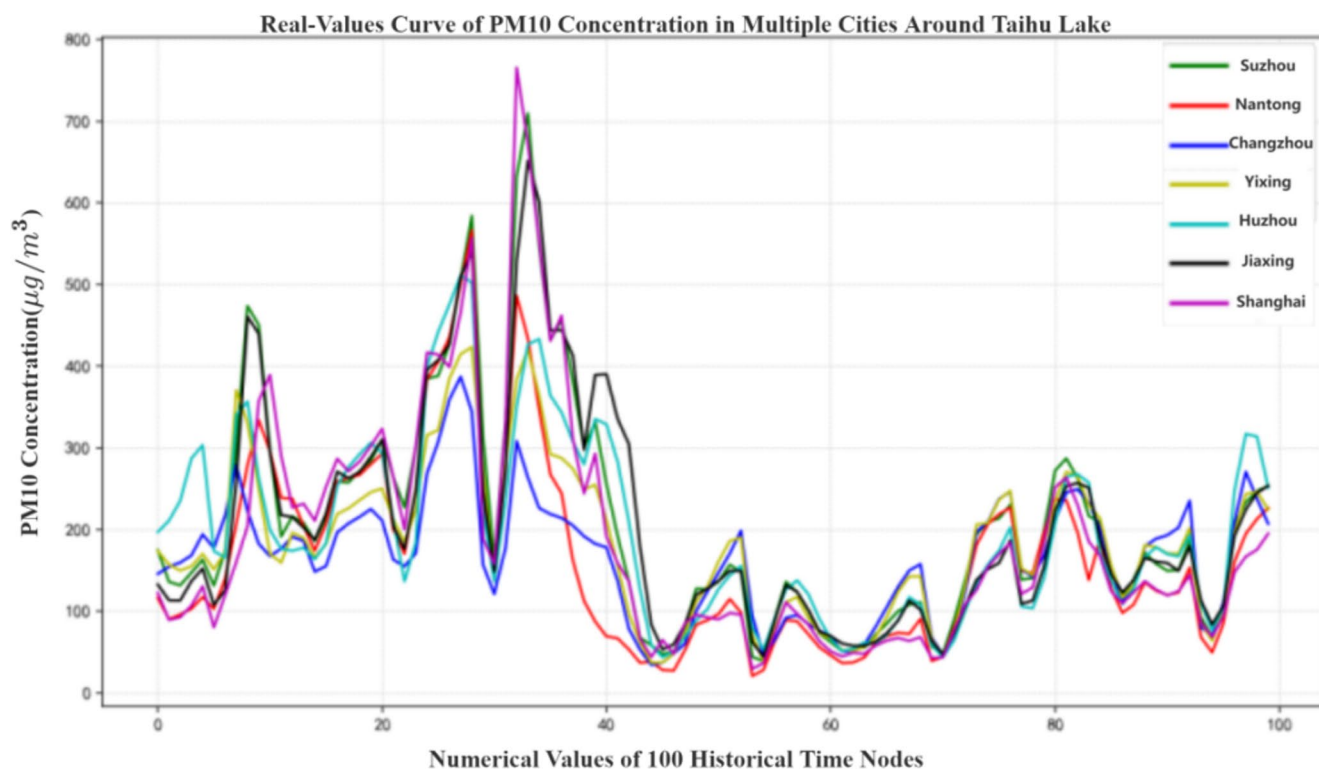


Fig. 16 Sample of PM10 concentration data for cities surrounding Taihu Lake

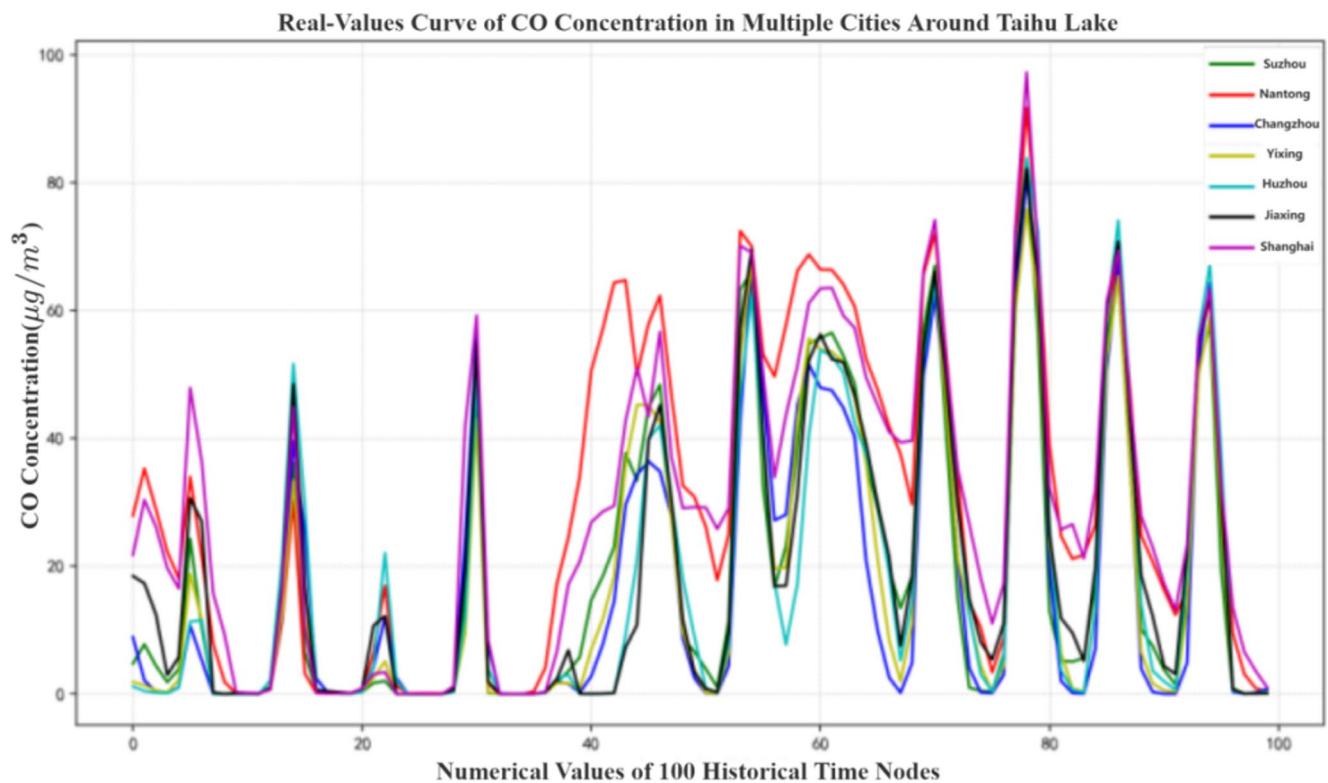


Fig. 17 Sample of CO concentration data for cities surrounding Taihu Lake

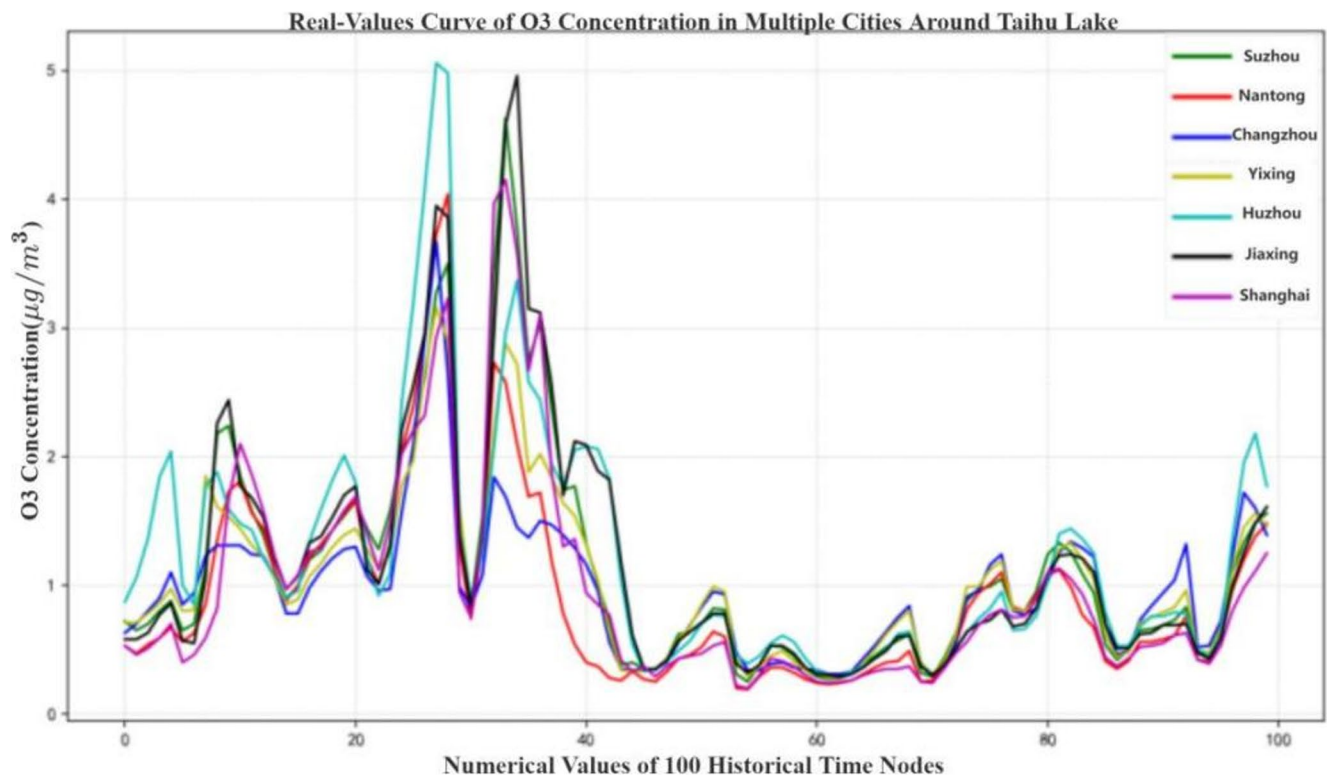


Fig. 18 Sample of O₃ concentration data for cities surrounding Taihu Lake

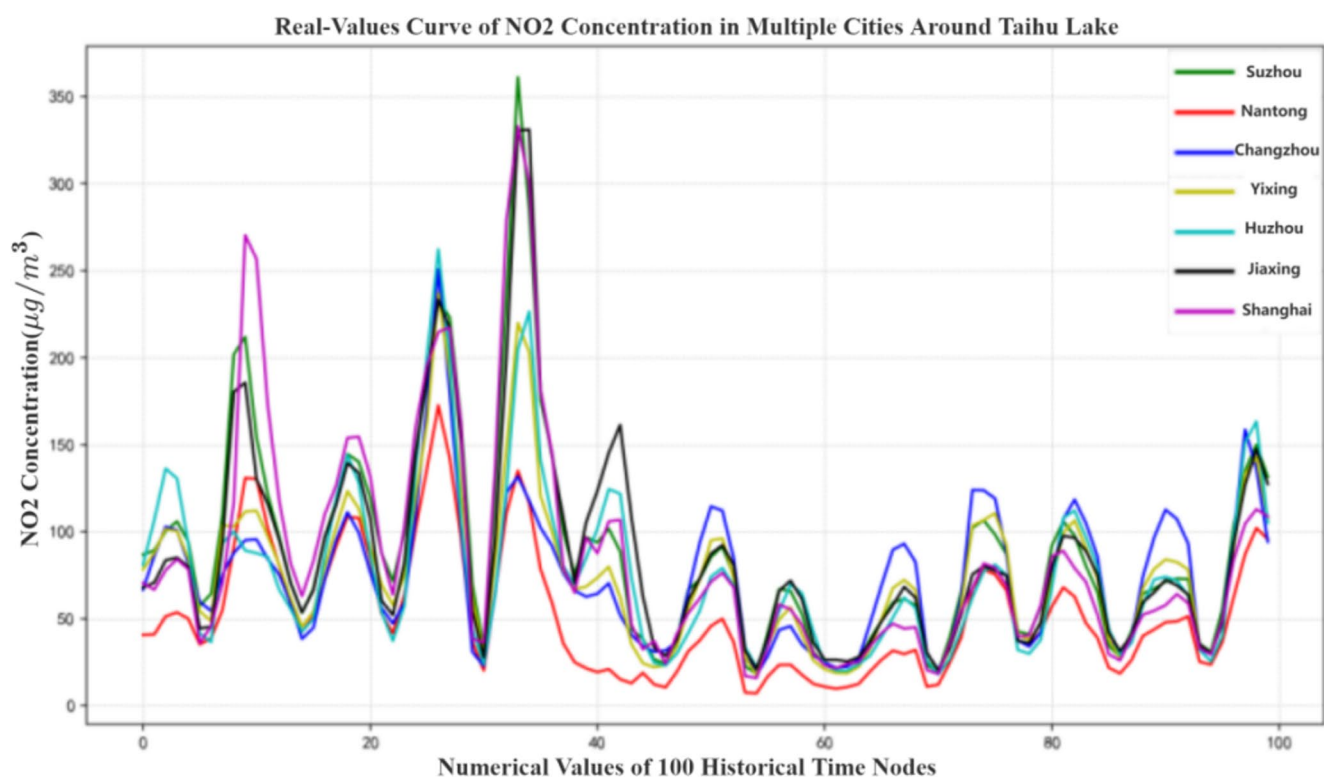


Fig. 19 Sample of NO₂ concentration data for cities surrounding Taihu Lake

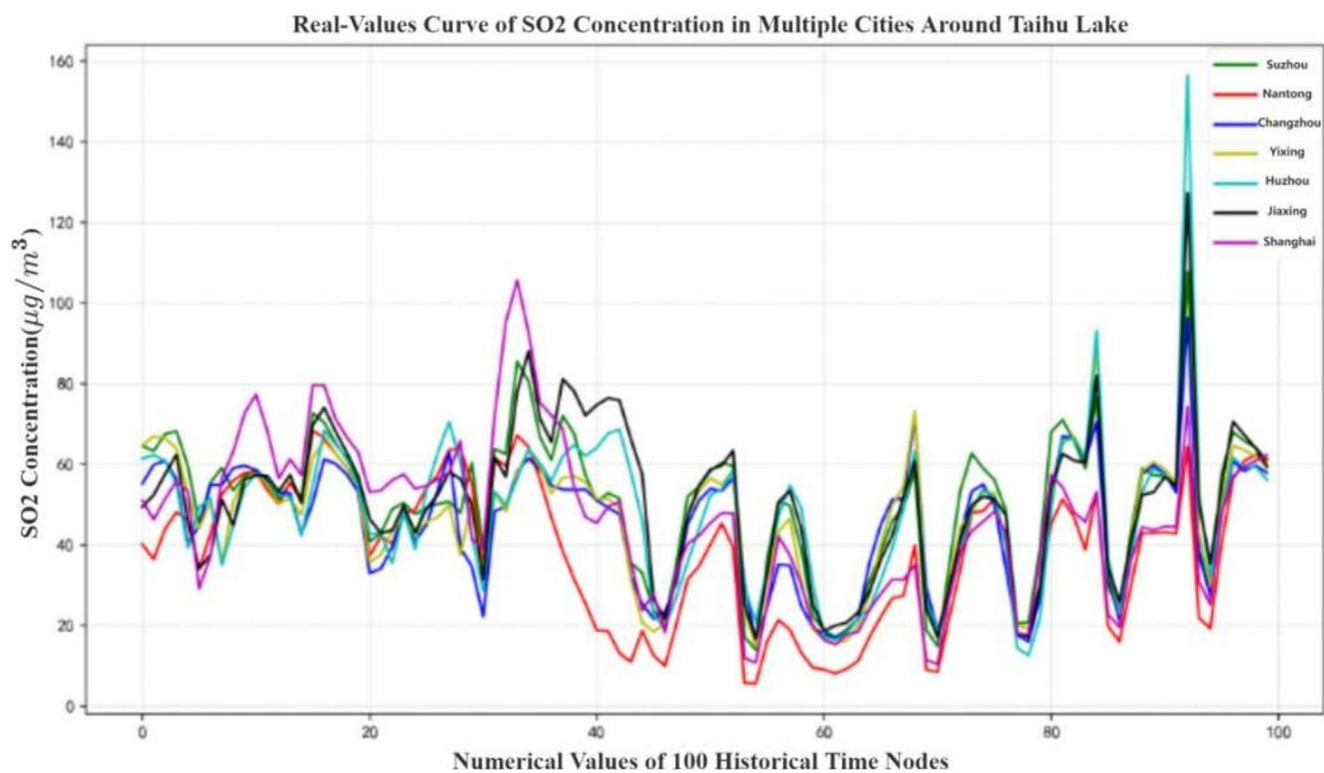


Fig. 20 Sample of SO₂ concentration data for cities surrounding Taihu Lake

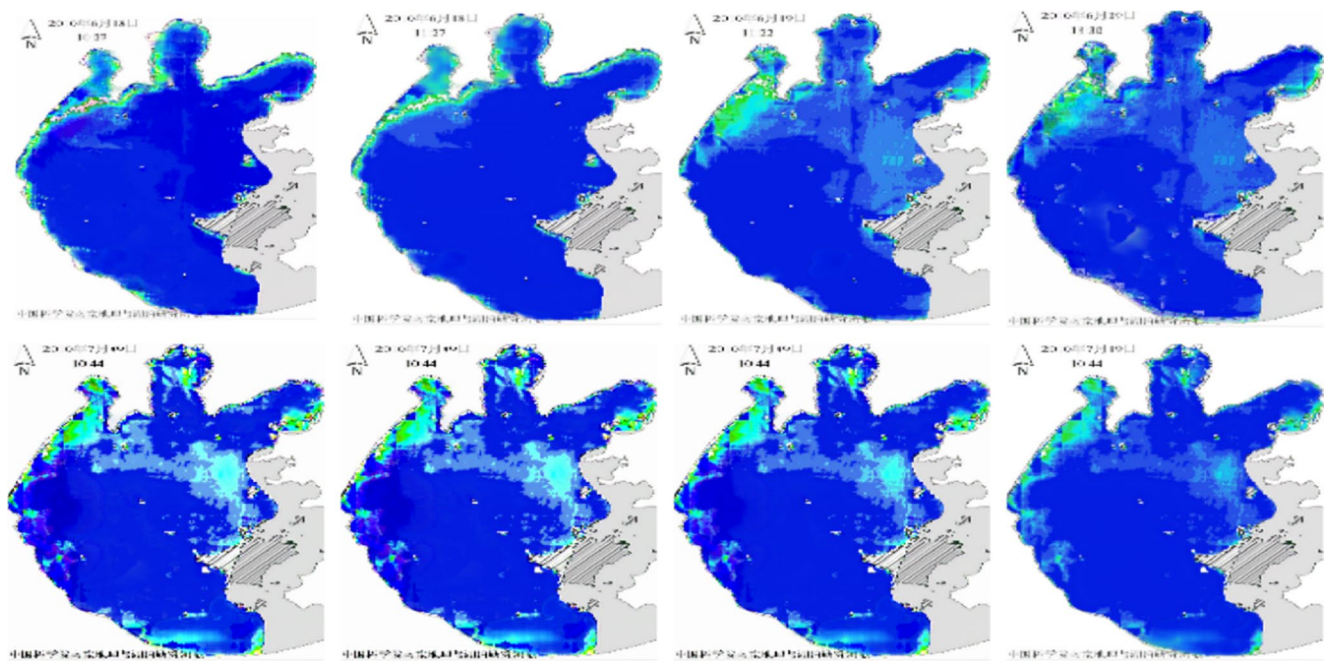


Fig. 21 Sample of the expanded Taihu Lake chlorophyll remote sensing image data

Table 2 Example of missing data in experimental datasets

Time	T1	T2	T3	T4	T5	T6	T7	T8
PM2.5 Image	I1	I2	I3	I4	I5	I6	I7	I8
Remote Sensing Image	Water1	Missing	Water3	Water4	Missing	Water6	Missing	Water8

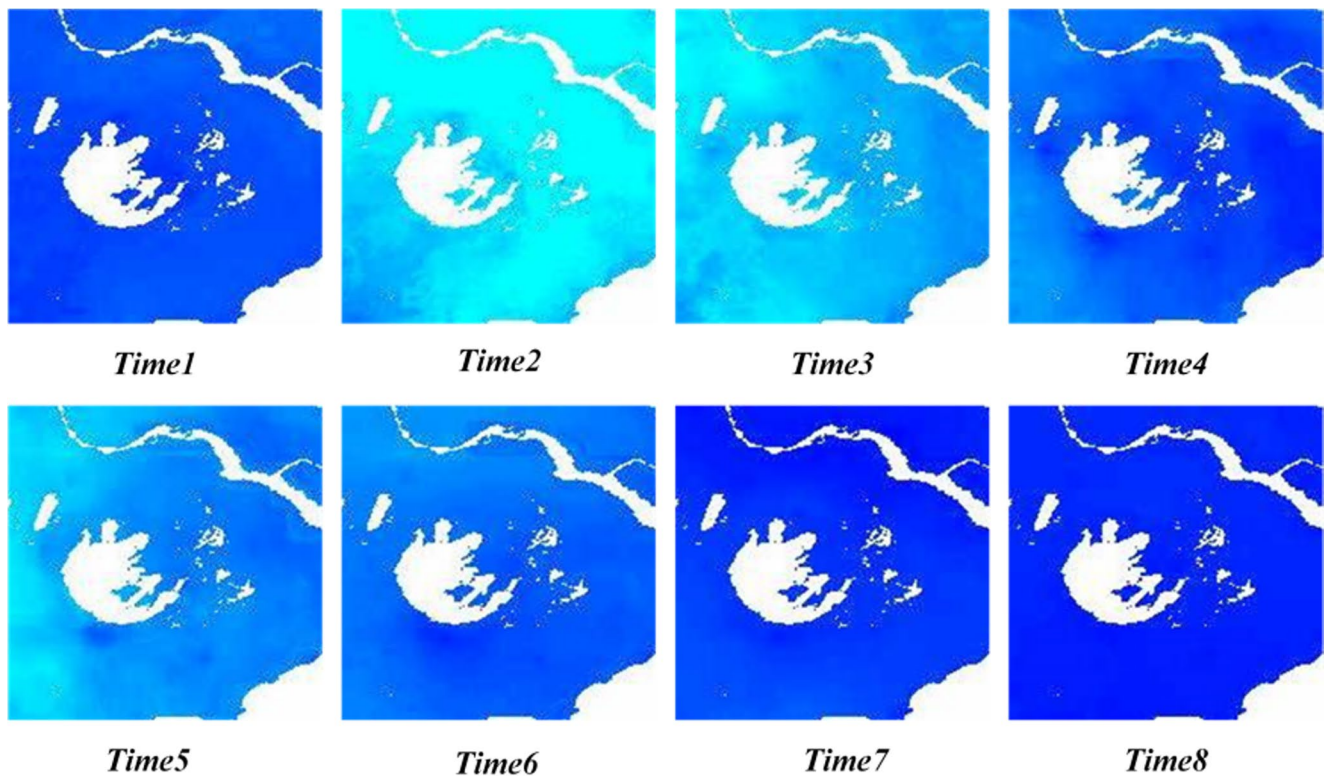


Fig. 22 Example of Yangtze River Delta PM2.5 time series image data

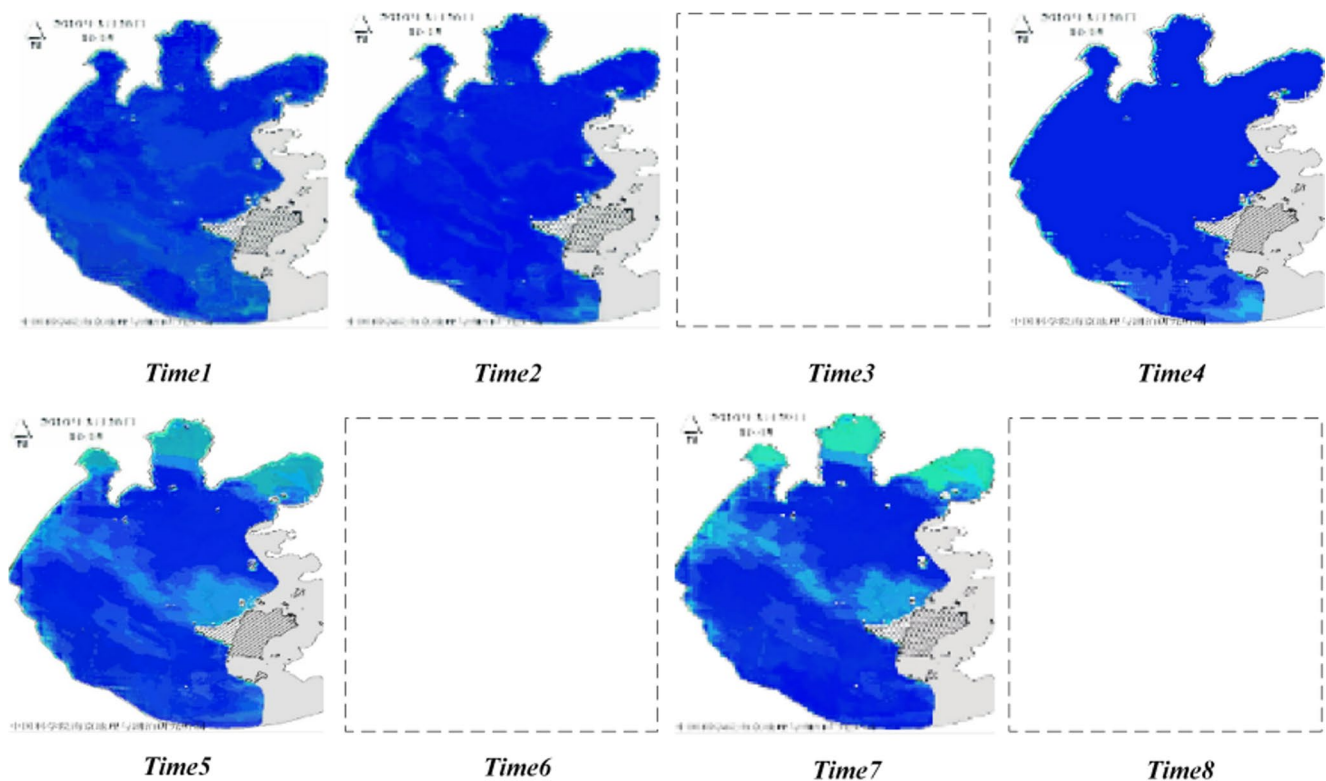


Fig. 23 Example of Taihu Lake Chlorophyll inversion time series image data

Table 3 Experimental Process of TL-MCAS_CSG_GAN Model

Step	Operation	Figure
Step1	Select the images corresponding to the missing time points in the Taihu Lake chlorophyll image dataset from the Yangtze River Delta PM2.5 high-resolution image dataset (e.g., Time3, Time6, and Time8) and organize them into a separate dataset as the input data for the TL-MCAS_CSG_GAN model	Figs. 22, 23, and 24
Step2	The TL-MCAS_CSG_GAN model learns from the dataset in Step 1 to generate preliminary Taihu Lake chlorophyll images for the missing time points	Figs. 24 and 25
Step3	Organize the preliminary generated Taihu Lake chlorophyll images for the missing time points and the original Taihu Lake chlorophyll images into a separate dataset as the input data for the knowledge transfer module	Fig. 26
Step4	The knowledge transfer module learns from the dataset in Step 3 to generate the final Taihu Lake chlorophyll images for the missing time points	Fig. 27

the MCAS_CSG_GAN model are 0.65424, 20.56574, and 0.92568, respectively, representing improvements of 0.01723, 0.65717, and 0.02615 compared to the CSG_GAN model, and improvements of 0.01354, 0.42523, and 0.02005 compared to the MCAS_GAN model, with average improvement percentages of 3.11% and 2.16%, respectively. It can be concluded that combining the CSG unit

and the MCAS unit, compared to using them individually, makes the model structure more flexible, allowing units to be added or removed based on actual needs, improving the model's performance on specific tasks. Additionally, since each unit can be trained and optimized independently, computation can be performed more efficiently, reducing computational costs. The introduction of the multiscale spatial pooling unit and the multichannel attention selection unit can enhance the model's feature extraction capability, robustness, efficiency, and performance, while reducing the risk of overfitting and enhancing feature expression, making the model more effective and reliable in handling complex tasks and various challenges.

Comparing the three evaluation metrics of the MCAS_CSG_GAN model and the TL-MCAS_CSG_GAN model, the SSIM, PSNR, and COSIN values of the TL-MCAS_CSG_GAN model are 0.69548, 21.84144, and 0.97623, respectively, representing improvements of 0.04124, 1.2757, and 0.05055, with an average improvement percentage of 6.27%. It can be concluded that transfer learning plays a significant role and has advantages in image generation models, helping researchers more effectively utilize existing models and data, improving generation quality, reducing computational costs, and achieving high-quality image generation with limited data and computational resources.

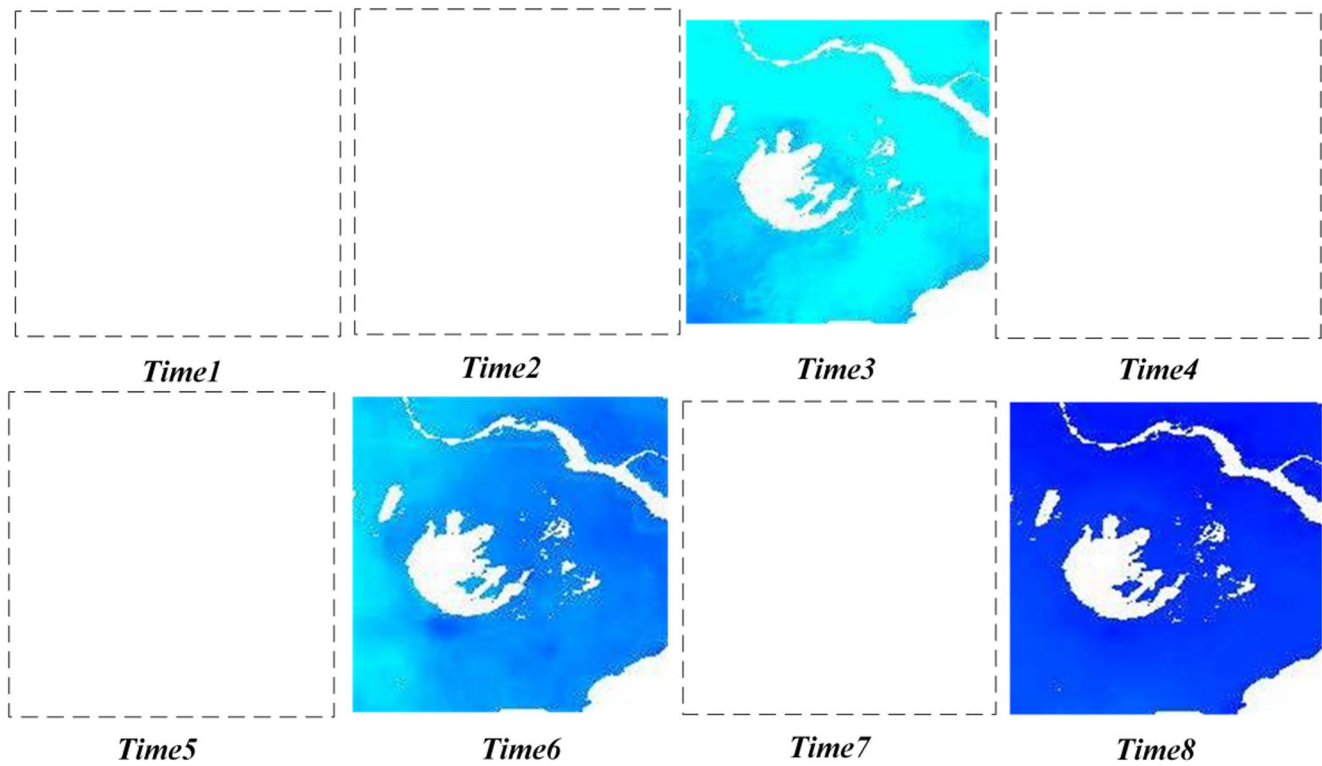


Fig. 24 Example of Yangtze River Delta PM2.5 images at missing time points (Time3, 6, 8)

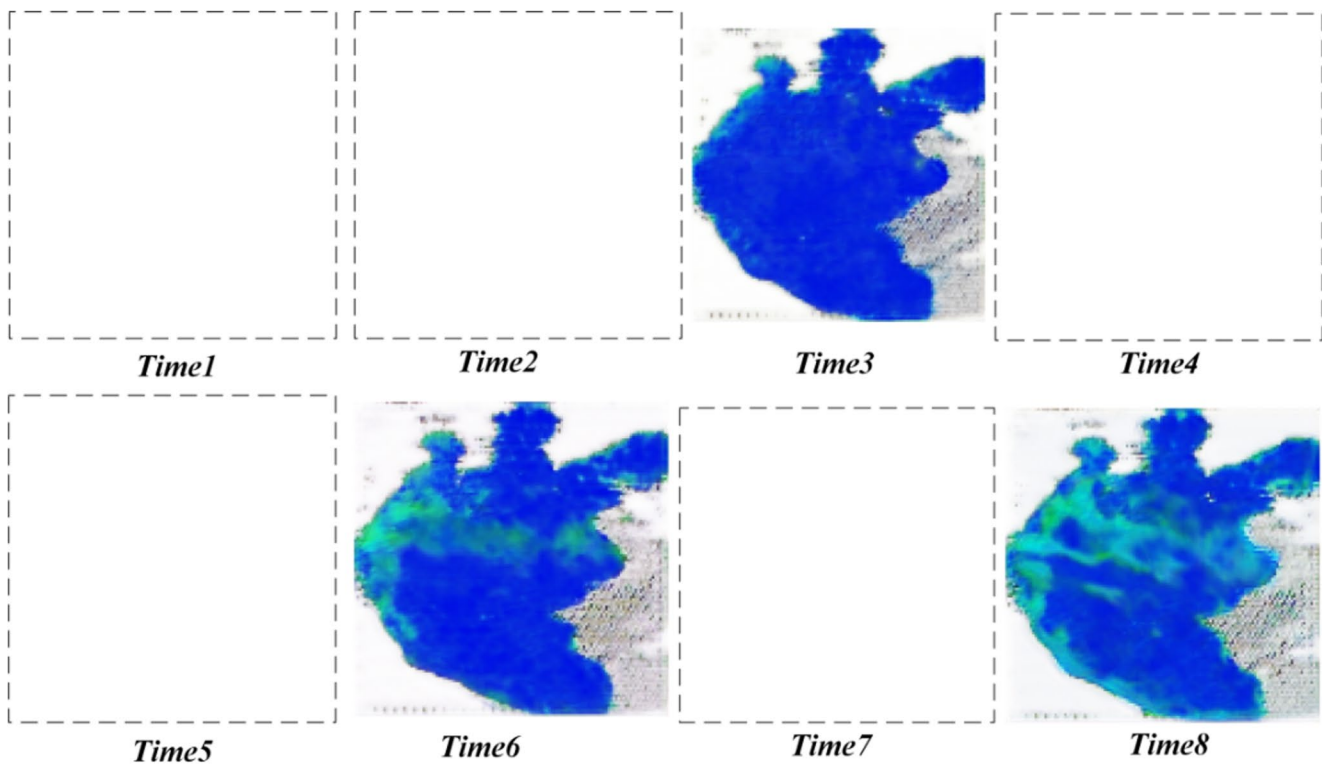


Fig. 25 Example of preliminary generated Taihu Lake Chlorophyll images at missing time points (Time3, 6, 8)

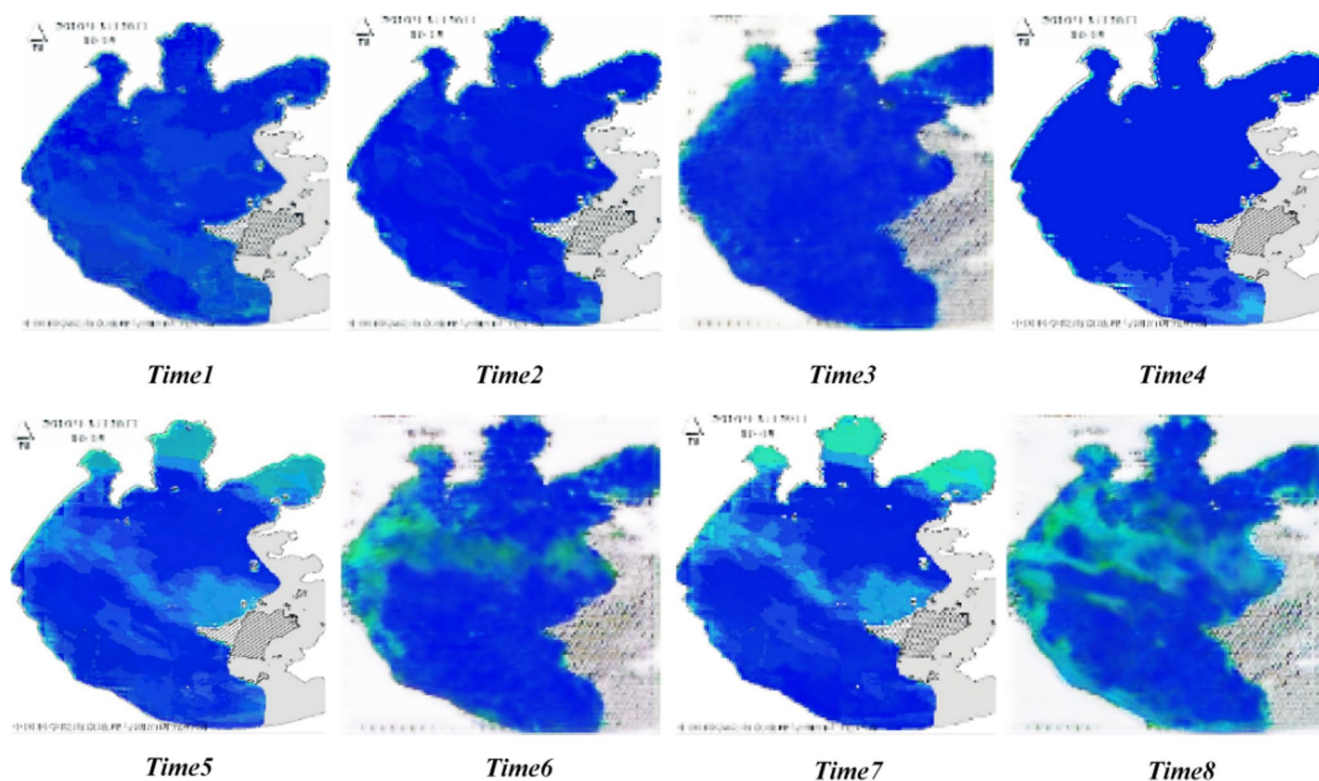


Fig. 26 Example of integration of preliminary generated Taihu Lake Chlorophyll images at missing time points (Time3, 6, 8) and Original Images

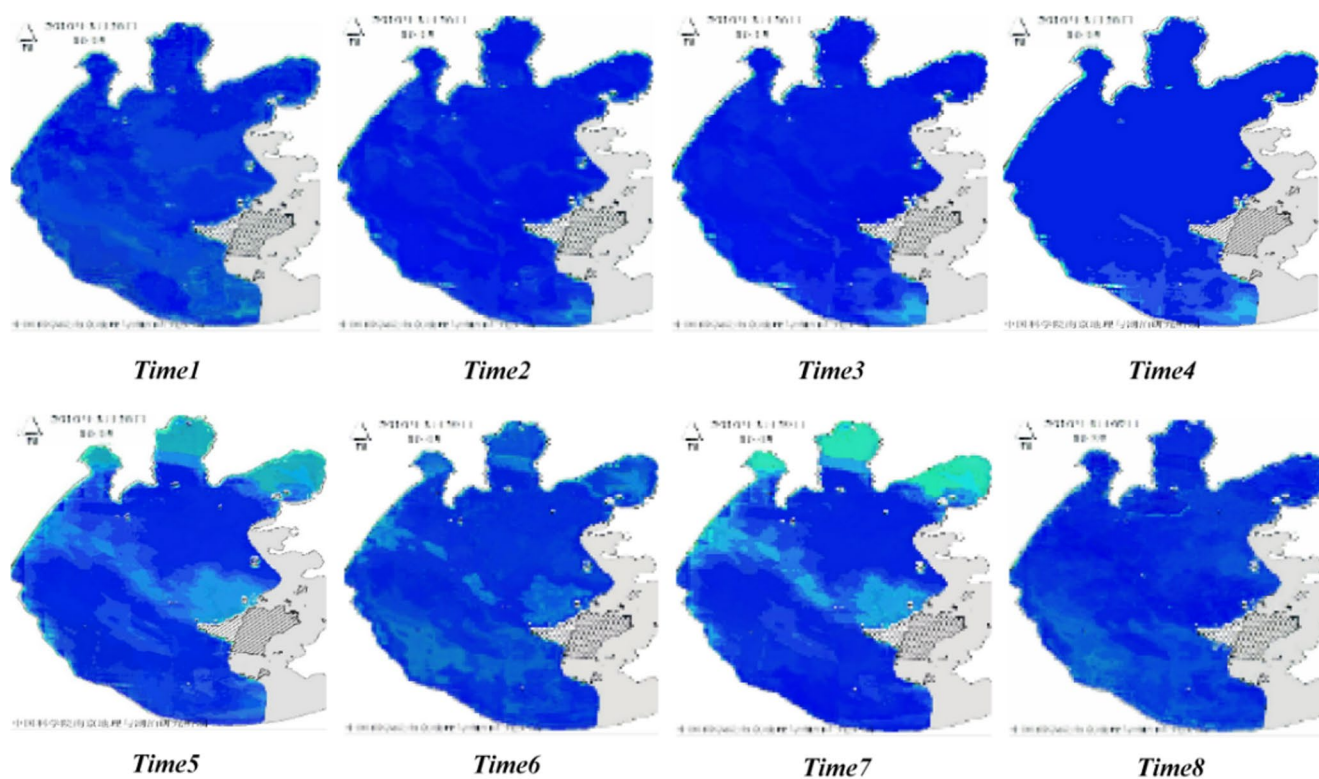


Fig. 27 Example of experimental results of TL-MCAS_CSG_GAN model

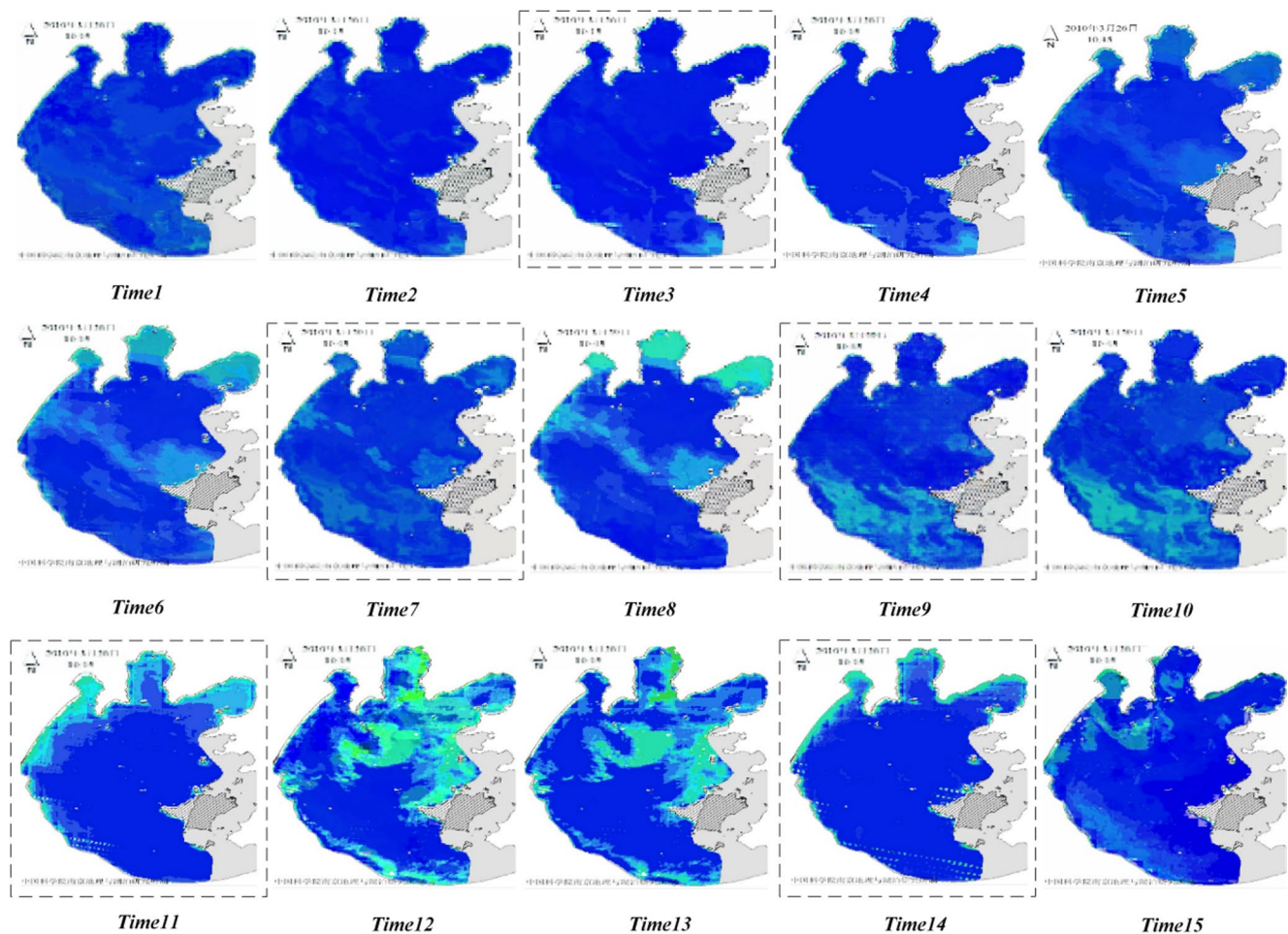


Fig. 28 Complete Time Series of Taihu Lake Chlorophyll Remote Sensing Image Data

Construction and prediction results of the atmospheric-water quality multimodal fusion prediction model

This section uses the Zhixing Cloud computing platform to support experimental calculations, selecting the Windows 10 operating system, equipped with a single-core GPU, graphics card type GeForce RTX 2080 Ti (11GB video memory), system image win10 cuda11, system disk size 350GB, and memory size starting from 28GB. The deep learning environment is configured with cuda11.1, cuDnn8, Anaconda, Pytorch1.7.1, Tensorflow2.4, and python3.8.3. In network training, the initial learning rate is set to 0.001, the batch size is set to 64, and the overall training cycle is set to 300 epochs. Additionally, the Adam optimizer is selected for its adaptive learning rate adjustment characteristics, which helps the model optimize the loss function more effectively during training. The specific network layer parameters of the Att_Conv2D_LSTM module are shown in Table 5, and the specific network layer parameters of

the Taihu Lake chlorophyll remote sensing image feature decoder are shown in Table 6.

This paper constructs a discriminator as a Markov discriminator model, and its detailed network parameters are shown in Table 7. The Markov discriminator structure consists of four network layers, aiming to extract and judge features at different scales in the input image.

During the training process of this network, this paper uses historical Taihu Lake chlorophyll remote sensing images from 30 time points as input for each batch. These images contain rich spatiotemporal information, helping the model capture the dynamic changes of Taihu Lake chlorophyll remote sensing images. By learning from these historical images, the model can predict the Taihu Lake chlorophyll remote sensing image at the 31st time point, achieving effective prediction of future Taihu Lake chlorophyll remote sensing images. The structural characteristics of the Markov discriminator make it focus on local details of remote sensing images, improving prediction accuracy and robustness.

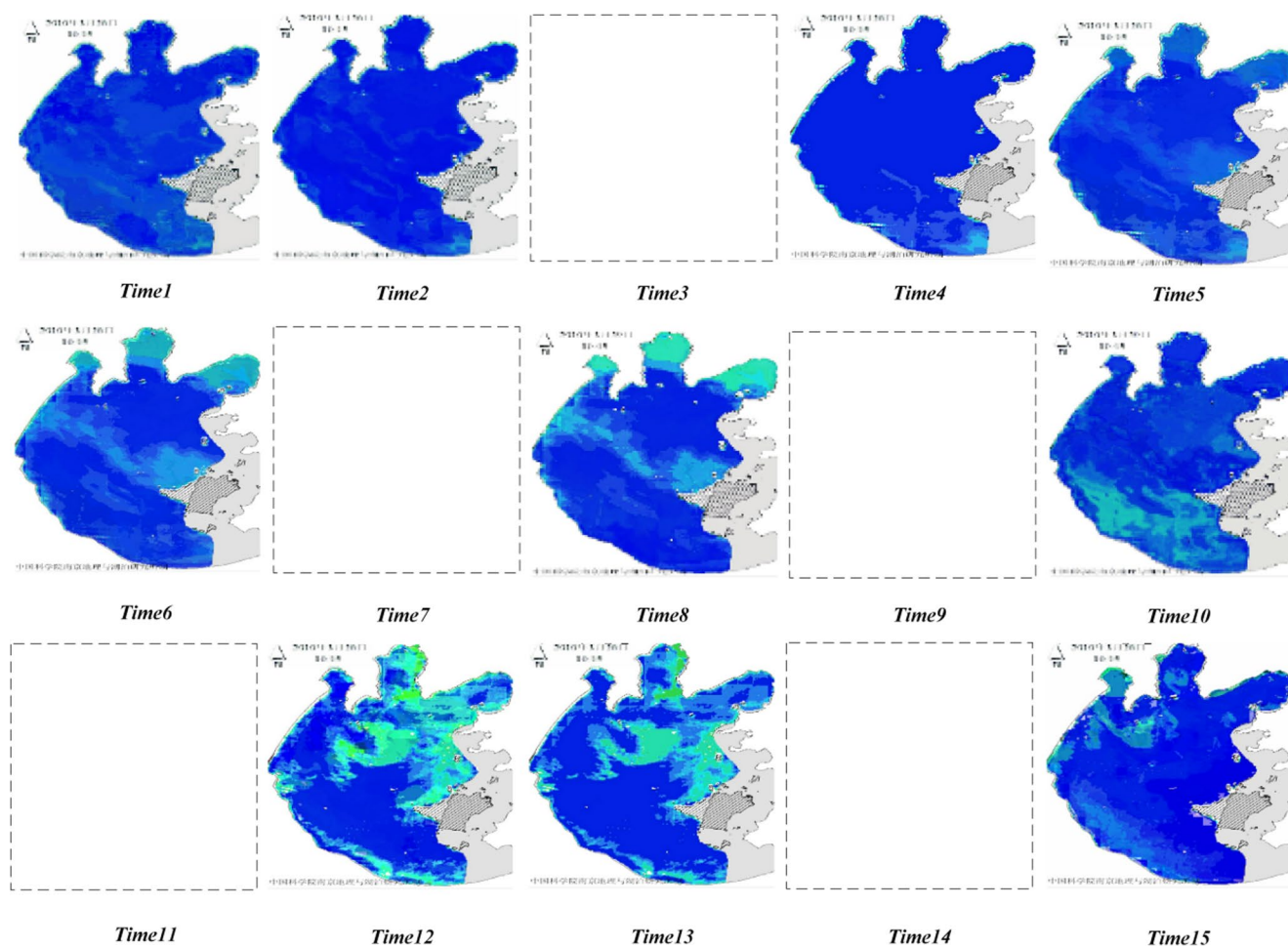


Fig. 29 Manually Missing Time Series of Taihu Lake Chlorophyll Remote Sensing Image Data

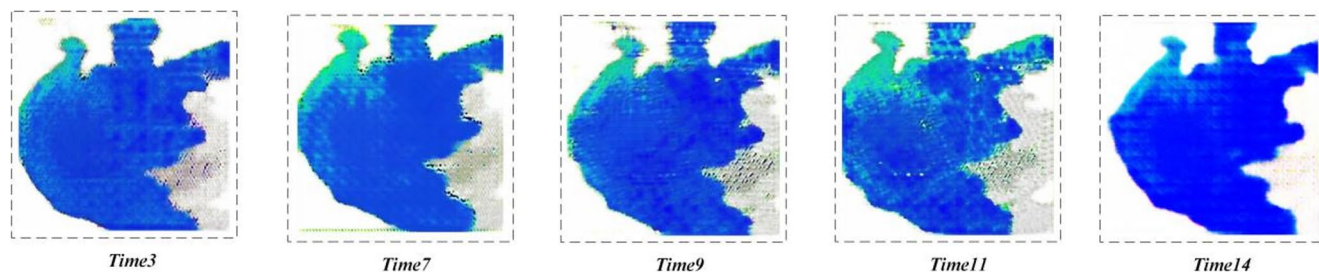


Fig. 30 Expanded Time Series of Taihu Lake Chlorophyll Remote Sensing Image Data Based on GAN Model

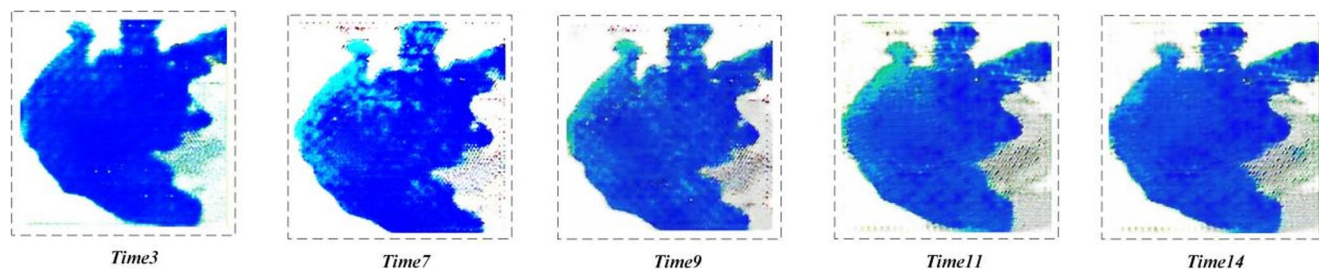


Fig. 31 Expanded Time Series of Taihu Lake Chlorophyll Remote Sensing Image Data Based on CSG_GAN Model

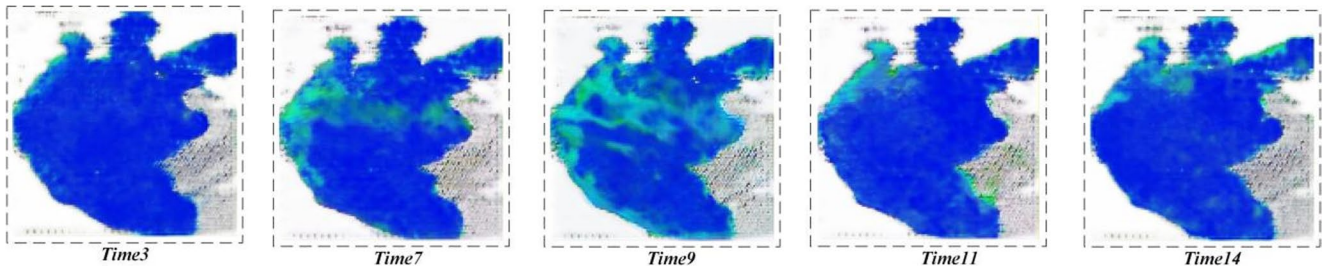


Fig. 32 Expanded Time Series of Taihu Lake Chlorophyll Remote Sensing Image Data Based on MCAS_GAN Model

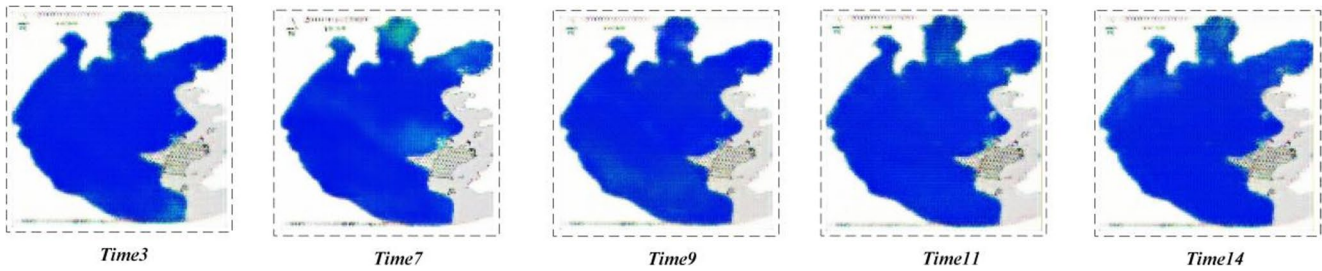


Fig. 33 Expanded Time Series of Taihu Lake Chlorophyll Remote Sensing Image Data Based on MCAS_CSG_GAN Model

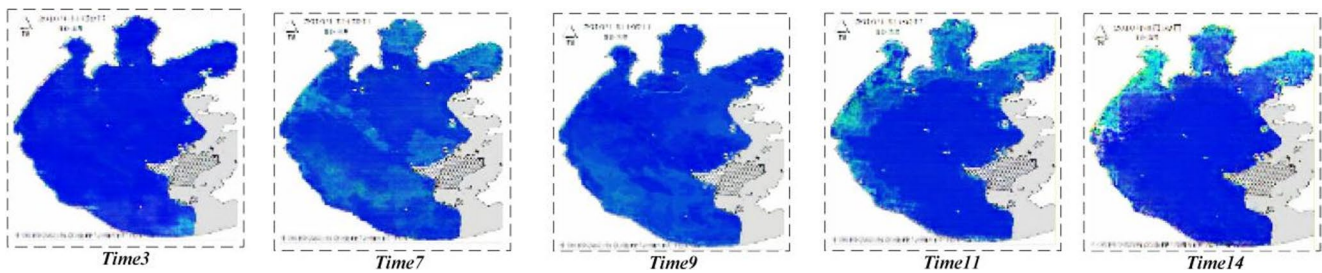


Fig. 34 Expanded Time Series of Taihu Lake Chlorophyll Remote Sensing Image Data Based on TL-MCAS_CSG_GAN Model

Table 4 Evaluation metrics for Taihu Lake Chlorophyll remote sensing image data expansion experiment

Model	SSIM	PSNR	COSIN	Average Improvement Ratio
GAN	0.61547	19.32871	0.87164	0%
CSG_GAN	0.63701	19.90857	0.89953	3.12%
MCAS_GAN	0.64070	20.14051	0.90563	4.07%
MCAS_CSG_GAN	0.65424	20.56574	0.92568	6.23%
TL-MCAS_CSG_GAN	0.70102	21.84144	0.97623	12.51%

Table 5 Parameter Settings of Att_Conv2D_LSTM

Network Layer	Parameters	Activation Function
Attention Average Pooling Layer	Outputsize=1	None
Spatial Attention Convolutional Layer	Filters=1, Kernelsize=7	Sigmoid
Channel Attention Fully Connected Layer	Outputfeatures=8	Sigmoid
Conv2D Network Layer	Filters=32, Kernelsize=4	Leaky-ReLU
LSTM Network Layer	Filters=32, Kernelsize=4	Leaky-ReLU

Table 6 Parameter Settings of Taihu Lake Chlorophyll Remote Sensing Image Feature Decoder

Network Layer	Parameters	Activation Function
ReflectionPad2d	inputsizes=3	None
Conv2D_1	Filters=1, Kernelsize=7	ReLU
Conv2D_2	Filters=1, Kernelsize=3	ReLU
ReflectionPad2d	inputsizes=3	None
Conv2D_3	Filters=1, Kernelsize=7	Tanh

Table 7 Parameter Settings of Markov Discriminator

Network Layer	Number of Convolutional Kernels	Convolutional Kernel Size	Activation Function
Conv2D-1	32	4×4	LeakyReLU
Conv2D-2	64	4×4	LeakyReLU
Conv2D-3	128	4×4	Sigmoid
Output Layer	1	4×4	None

To verify that the proposed atmospheric-water quality multimodal fusion water eutrophication prediction model based on cross-attention has good prediction effects, this section compares the GAN, Conv2D_LSTM_GAN,

Att_Conv2D_LSTM_GAN, and Att_Conv2D_LSTM_Cross_GAN (the proposed model in this study) models in predicting Taihu Lake chlorophyll remote sensing images. The specific comparison results are shown in Fig. 35.

To objectively evaluate the results of the ablation experiment, Table 8 shows the average SSIM, PSNR, and COSIN metrics for each part of the ablation experiment, as well as

the improvement effects, verifying that the introduction of the Conv2D_LSTM module, Attention module, and cross-attention mechanism for fusing atmospheric six-parameter factors significantly improves the model's prediction performance.

The purpose of the experiment is to verify that the proposed Att_Conv2D_LSTM_Cross_GAN model is the most

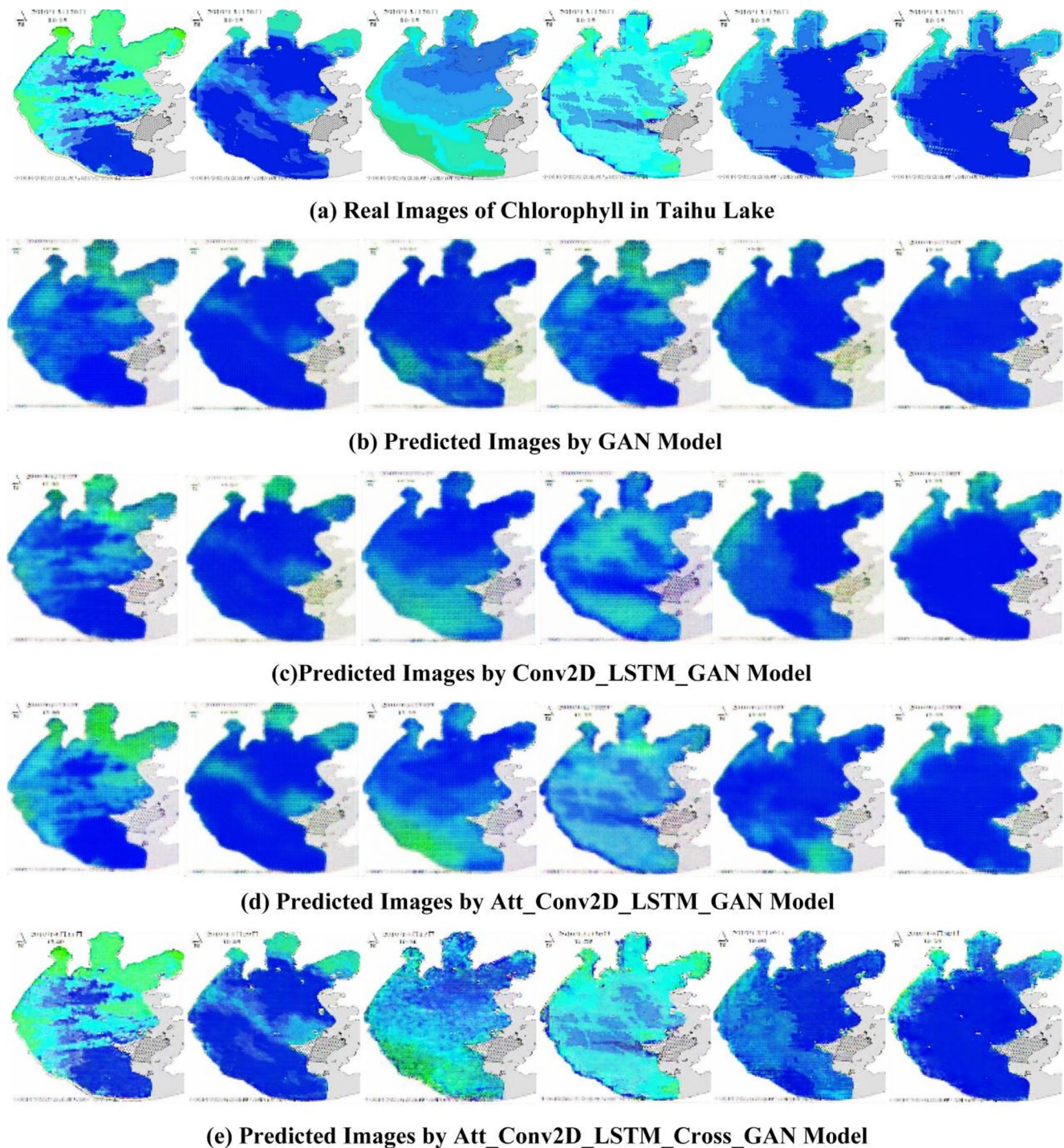


Fig. 35 Comparison of Ablation Experiment Results for Taihu Lake Chlorophyll Remote Sensing Image Prediction

Table 8 Evaluation metrics of ablation experiment for Att_Conv2D_LSTM_Cross_GAN model

Model	SSIM	PSNR	COSIN	Improvement Ratio
GAN	0.69012	23.71583	0.91538	0%
Conv2D_LSTM_GAN	0.70619	24.28500	0.94284	2.33%
Att_Conv2D_LSTM_GAN	0.71406	24.5387	0.94741	3.47%
Att_Conv2D_LSTM_Cross_GAN	0.73504	25.37593	0.97828	6.51%
Mean-30	0.70912	22.93215	0.94422	
LinearTrend-30	0.69693	21.93421	0.93304	

effective model for predicting Taihu Lake chlorophyll remote sensing images compared to the Att_Conv2D_LSTM_GAN, Conv2D_LSTM_GAN, and GAN models (note: the three comparison models did not add the cross-attention mechanism Cross, i.e., they did not fuse atmospheric six-parameter factors from multiple cities around Taihu Lake for modeling and prediction). By comparing the prediction results of each model in Fig. 35 and analyzing the evaluation metrics of each model in Table 8, the following conclusions can be drawn:

As shown in Table 8, the prediction performance improves progressively as additional modules are incorporated into the framework. To better interpret these quantitative results, the contribution of each component is analyzed from three aspects: (1) temporal dependency modeling, (2) attention-based feature refinement, and (3) atmospheric multimodal feature fusion.

First, compared with the baseline GAN model, the Conv2D_LSTM_GAN improves SSIM from 0.69012 to 0.70619 and PSNR from 23.72 dB to 24.29 dB. This improvement demonstrates that introducing the LSTM module enables the model to capture temporal dependencies in chlorophyll evolution, thereby enhancing prediction stability and structural consistency.

Second, when the attention mechanism is incorporated (Att_Conv2D_LSTM_GAN), SSIM further increases to 0.71406 and PSNR to 24.54 dB. This indicates that attention-based feature weighting allows the model to emphasize spatially informative regions and suppress irrelevant variations, resulting in refined prediction performance.

Third, the most notable performance gain is observed when atmospheric six-parameter features are integrated through the cross-attention mechanism. Compared with the Att_Conv2D_LSTM_GAN model (without atmospheric inputs), the proposed Att_Conv2D_LSTM_Cross_GAN achieves improvements of $\Delta\text{SSIM}=+0.02098$, $\Delta\text{PSNR}=+0.83723$ dB, and $\Delta\text{COSIN}=+0.03087$, corresponding to an overall improvement ratio of 6.51%. Since the primary architectural difference between these two models lies in the inclusion of atmospheric pollutant

features, the observed performance enhancement can be attributed to multimodal information fusion. These findings suggest that atmospheric temporal dynamics provide complementary spatiotemporal signals for chlorophyll-a prediction and contribute to improved eutrophication modeling accuracy.

In addition to GAN-based models, two classical statistical baselines, Mean-30 and LinearTrend-30, were introduced for comparison. The Mean-30 model predicts the chlorophyll image at time $t+1$ by averaging the previous 30 observations, assuming temporal stability. The LinearTrend-30 model performs pixel-wise linear regression over the past 30 days and extrapolates the trend to predict the next day.

As shown in Table 8, although the Mean-30 baseline achieves relatively competitive SSIM (0.70912) and COSIN (0.94422), its PSNR (22.93 dB) remains significantly lower than that of the proposed multimodal fusion model. This indicates that simple temporal averaging can preserve global structural similarity but fails to capture fine-grained spatial variations.

The LinearTrend-30 baseline performs slightly worse than Mean-30 across all metrics, suggesting that linear temporal extrapolation is insufficient to model the nonlinear evolution of chlorophyll dynamics.

Compared with both statistical baselines, the proposed Att_Conv2D_LSTM_Cross_GAN consistently achieves higher SSIM, PSNR, and COSIN values, demonstrating its superior ability to model complex spatiotemporal patterns and multimodal interactions.

Water eutrophication evaluation

Chlorophyll is the main parameter characterizing water eutrophication. The following is the Carlson evaluation index formula for converting chlorophyll as a single water quality parameter indicator into water eutrophication degree:

$$TSI(Chl.a) = 10(2.5 + 1.086 \ln(Chl.a)) \quad (4.1)$$

In the formula, TSI represents the Carlson Trophic State Index, and $Chl.a$ represents the chlorophyll concentration in the water body, with the unit: $\mu\text{g}/\text{L}$. According to the correspondence between the TSI index and the trophic state of the water body, a series of continuous numbers from 0 to 100 are used to divide the trophic state into five levels: oligotrophic ($TSI < 35$), mesotrophic ($35 \leq TSI < 50$), mildly eutrophic ($50 \leq TSI < 60$), moderately eutrophic ($60 \leq TSI < 70$), and hypereutrophic ($TSI \geq 70$). The corresponding chlorophyll concentration indicator levels are shown in Fig. 36.

Chlorophyll Concentration		Nutrient Status
	0.0-2.5ug/L	Oligotrophic
	2.5-10.0ug/L	Mesotrophic
	10.0-25.0ug/L	Mildly Eutrophic
	25.0-40.0ug/L	Mildly Eutrophic
	40.0-50.0ug/L	Moderately Eutrophic
	50.0-65.0ug/L	Moderately Eutrophic
	>65.0ug/L	Severely Eutrophic

Fig. 36 Chlorophyll concentration corresponding to water eutrophication state levels

By displaying the true images, predicted images, and water eutrophication state levels of chlorophyll remote sensing at two time points, as shown in Fig. 37, the atmospheric-water quality multimodal information fusion water

eutrophication prediction is analyzed. Comparing the true and predicted images at time (a), it can be seen that the water eutrophication level in the middle and northeast part of the water body is oligotrophic in both the true and predicted results, while the water eutrophication level in the middle and southwest part of the water body has errors, with the true result being mildly eutrophic and the predicted result being between mesotrophic and mildly eutrophic. Comparing the true and predicted images at time (b), it can be seen that the overall true and predicted results are oligotrophic, with small errors in the water eutrophication level along the northwest and northeast edges of the lake, where the true result is moderately eutrophic and the predicted result is between mildly eutrophic and moderately eutrophic.

To verify the accuracy of the water eutrophication evaluation results in this paper, the annual average chlorophyll remote sensing images are compared with the data published in the China Lake Ecological Environment Research Report - Taihu Lake. The results are shown in Table 9.

Overall, the predicted TSI values show small numerical deviations from the observed data across all years, indicating

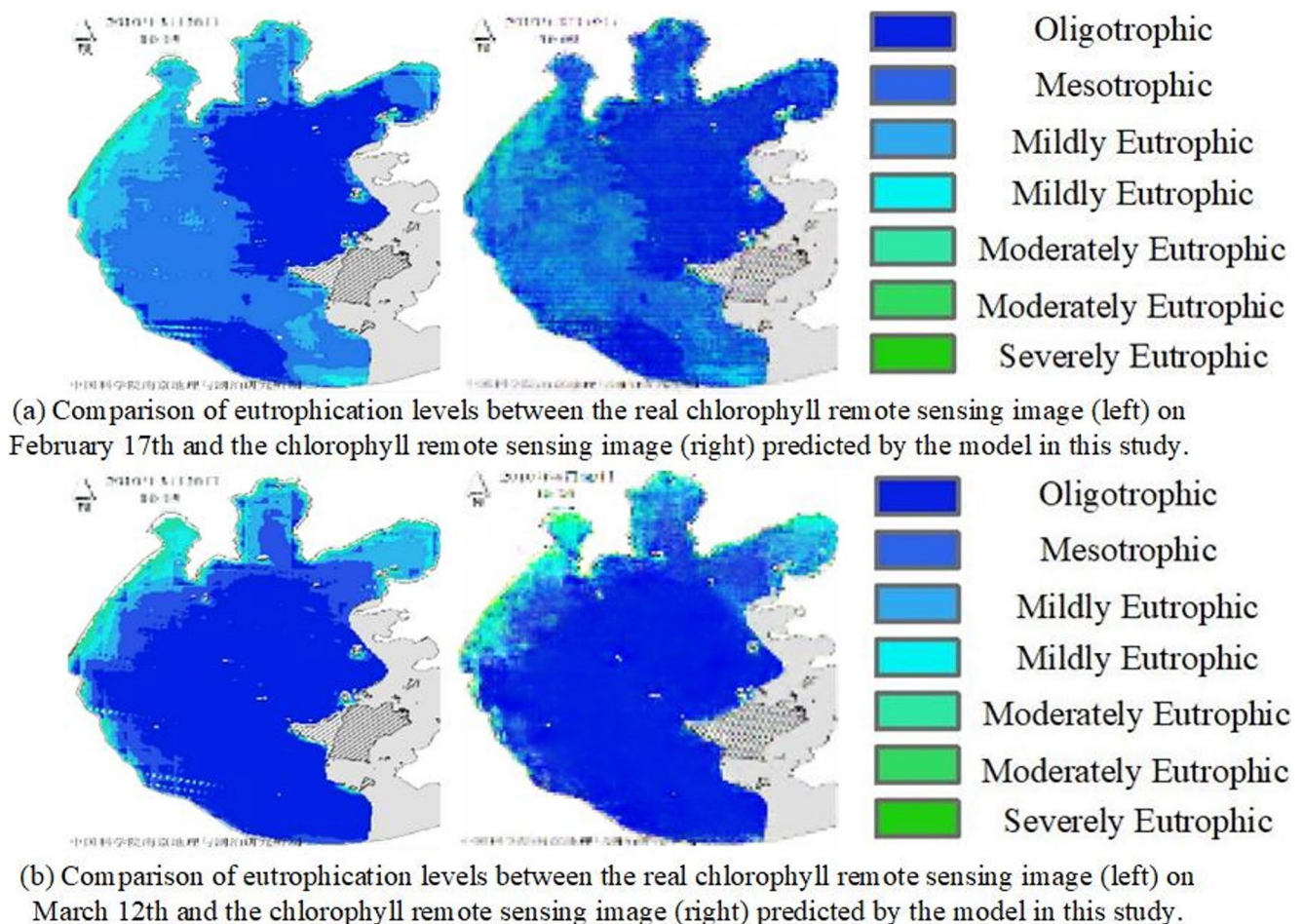


Fig. 37 Comparison of Eutrophication Levels Between True and Predicted Results of Chlorophyll Remote Sensing Images at Two Time Points

Table 9 Evaluation of Taihu Lake eutrophication level

Lake	Year	Predicted TSI Index	Predicted Trophic Level	True TSI Index	True Trophic Level
Taihu Lake	2014	55.6	Mildly Eutrophic	57.8	Mildly Eutrophic
Taihu Lake	2015	59.2	Mildly Eutrophic	60.7	Moderately Eutrophic
Taihu Lake	2016	59.6	Mildly Eutrophic	58.8	Mildly Eutrophic
Taihu Lake	2017	62.3	Moderately Eutrophic	60.9	Moderately Eutrophic

stable regression performance and consistent eutrophication trend estimation. The classification discrepancy observed in 2015 is primarily due to the threshold-based categorization of trophic levels. The predicted TSI (59.2) is close to the classification boundary (TSI=60), while the observed value (60.7) lies slightly above this threshold. The absolute numerical difference is only 1.5 units, suggesting that the category shift is mainly caused by boundary sensitivity rather than substantial prediction error.

To further assess the physical consistency of the estimated chlorophyll-a values, we compared the lake-wide monthly mean concentrations of Taihu Lake with independent monitoring statistics (by Bulletin on Water Quality of Water Bodies at Provincial Boundaries in the Taihu Lake Basin and Southeastern Rivers). The predicted values for March (14.87 $\mu\text{g/L}$) and April (29.06 $\mu\text{g/L}$) show strong numerical agreement with the reported monthly means (14.70 $\mu\text{g/L}$ and 31.50 $\mu\text{g/L}$, respectively), with only minor deviations. This external comparison supports the physical plausibility of the derived chlorophyll-a estimates and their consistency with independent monitoring statistics.

Analysis and discussion

From the above prediction results, it can be seen that ablation experiments were conducted on both the proposed TL-MCAS_CSG_GAN and Att_Conv2D_LSTM_Cross_GAN models. By calculating the SSIM, PSNR, and COSIN metrics for the remote sensing images generated by the TL-MCAS_CSG_GAN model and the true images, the results show that the added cascaded semantic generation module, multichannel attention selection module, and knowledge transfer module all positively benefit the model, further verifying that the multichannel attention cascaded semantic selection module and knowledge transfer module can improve the accuracy of image transformation and generation. By comparing the water eutrophication prediction results of the Att_Conv2D_LSTM_Cross_GAN model with those of the model without atmospheric factors, it is verified that the proposed cross-modal feature encoder for fusing

atmospheric pollutant factors can improve the accuracy of water eutrophication prediction.

Furthermore, comparisons with classical statistical baselines (Mean-30 and LinearTrend-30) demonstrate that deep multimodal modeling significantly outperforms simple temporal averaging and linear extrapolation strategies, highlighting the necessity of nonlinear spatiotemporal representation learning for chlorophyll-a prediction.

Conclusion

This study proposes a two-stage multimodal fusion framework for water eutrophication prediction by integrating atmospheric pollutant information and chlorophyll remote sensing imagery.

First, a transfer learning-based chlorophyll image reconstruction strategy (TL-MCAS-CSG-GAN) is developed to address irregular temporal sampling and limited data availability. By improving temporal continuity and enriching training sequences, the reconstruction module enhances the stability of subsequent spatiotemporal modeling. It is emphasized that reconstructed samples are used solely for training purposes, while all quantitative evaluations are conducted exclusively on real chlorophyll observations to ensure experimental validity.

Second, a cross-attention Conv2D-LSTM GAN model is designed to fuse atmospheric six-parameter features with chlorophyll image sequences for eutrophication prediction. Experimental results demonstrate consistent improvements in SSIM, PSNR, and COSIN compared with both GAN-based variants and classical statistical temporal baselines. The ablation analysis further confirms that atmospheric pollutant information provides complementary spatiotemporal signals that contribute to enhanced prediction accuracy.

From an environmental informatics perspective, incorporating atmospheric pollutant factors offers a data-driven representation of external nutrient transport processes that may influence lake eutrophication dynamics. The proposed framework demonstrates the potential of multimodal environmental data integration for intelligent water quality monitoring.

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Author contributions Authorship Contribution Statement Li Wang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Project administration. Yihan Sun: Methodology, Validation, Investigation, Resources, Writing - original draft, Writing - review & editing. Xiaoyi Wang: Data curation, Visualization, Writing - review & editing. Jiping Xu: Validation, Writing - review & editing. Zhiyao Zhao: Software, Writing - review & editing. Jiabin Yu: Resources, Writing - review & editing. Huiyan Zhang: Formal analysis, Writing - review & editing. Qian Sun: Investigation, Writing - review & editing. Yuting Bai: Visualization, Writing - review & editing. All authors: Reviewed and approved the final manuscript.

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Data availability No datasets were generated or analysed during the current study.

Code availability Name of the code/library : water.
Contact: Yihan Sun email: sunyihan7895123@163.com.
Hardware requirements: This section uses the Zhixing Cloud computing platform to support experimental calculations, selecting the Windows 10 operating system, equipped with a dual-core GPU, graphics card type GeForce RTX 3090 (24GB video memory), system image win10 cuda11, system disk size 350GB, and memory size starting from 56GB. The deep learning environment is configured with cuda11.1, cuDnn8, Anaconda, Pytorch1.7.1, Tensorflow2.4, and python3.8.3.
Program language: python.
Software required: python.
Program size: pycharm.
The source codes are available for downloading at the link: <https://github.com/sunyihan123/water>.

Declarations

Competing interests The authors declare no competing interests.

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